

Program overview

19-May-2025 14:43

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Schakelprogramma Industrieel Ontwerpen	Bridging Programme Industrial Design Engineering		
Schakelprogramma Design for Interaction	Bridging Programme Design for Interaction		
Dfi schakelprogramma 1			
Design Focus			
IOB2-1-23	DP2: Designing Product Services	10	
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB4-B3-22	Codesign in Services	5	
Select one			
IOB4-P1	Human Factors & Ergonomics	5	
IOB4-P2	Experience, Motivation & Behaviour	5	
IOB4-P3	Culture & Society	5	
Dfi schakelprogramma 2			
Social/Interaction Focus			
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB4-1-24	DP4 Designing for Society	10	
Select two			
IOB4-B3-22	Codesign in Services	5	
IOB4-P1	Human Factors & Ergonomics	5	
IOB4-P2	Experience, Motivation & Behaviour	5	
IOB4-P3	Culture & Society	5	
IOB4-T3	Machine Learning for Design	5	
IOB6-E8	Design Analytics	5	
Schakelprogramma Industrial Product Design	Bridging Programme Industrial Product Design		
IPD schakelprogramma 1			
Design Focus			
IOB2-1-23	DP2: Designing Product Services	10	
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB4-B3-22	Codesign in Services	5	
Select one			
IOB4-P1	Human Factors & Ergonomics	5	
IOB4-P2	Experience, Motivation & Behaviour	5	
IPD schakelprogramma 2			
Technology/Engineering Focus			
IOB2-2-24	Digital Interfaces	5	
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB2-5-24	Intelligent Products	5	
IOB4-P1	Human Factors & Ergonomics	5	
Select one			
IOB4-T1-22	Materials and Manufacturing	5	
IOB4-T2	Mechatronics	5	
IOB4-T3	Machine Learning for Design	5	
IOB6-E8	Design Analytics	5	
Schakelprogramma Strategic Product Design	Bridging Programme Strategic Product Design		
SPD schakelprogramma 1			
Design Focus			
IOB2-1-23	DP2: Designing Product Services	10	
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
Select one			
IOB4-B1-22	Strategic Brand Management	5	
IOB4-B2	Business Model Design	5	
IOB4-B3-22	Codesign in Services	5	

IOB4-P1	Human Factors & Ergonomics	5	
IOB4-P2	Experience, Motivation & Behaviour	5	
IOB4-P3	Culture & Society	5	
IOB6-E8	Design Analytics	5	
SPD schakelprogramma 2			
Organizational Focus			
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB4-1-24	DP4 Designing for Society	10	
Select two			
IOB4-B1-22	Strategic Brand Management	5	
IOB4-B2	Business Model Design	5	
IOB4-B3-22	Codesign in Services	5	
IOB4-S2-24	Leadership and Project Organisation	5	
IOB4-S3	Mastering Research Methods	5	
IOB6-E8	Design Analytics	5	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Schakelprogramma Industrieel Ontwerpen

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Schakelprogramma Design for Interaction

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Dfi schakelprogramma 1

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Design Focus

IOB2-1-23	DP2: Designing Product Services	10
Course Coordinator	Ir. C.P.J.M. Kroon	
Contact Hours / Week x/x/x/x		
Education Period	3	
	4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Prerequisites		
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dp2-io@tudelft.nl	

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren. Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek.	
Study Goals	Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
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Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
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Special Information	Contact: uv-io@tudelft.nl	

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is structured in two pillars. In the first one, students learn through lectures, self-study, and group work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students also have lectures by external guests who provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. The second pillar of the course is about organising codesign sessions to tackle real-world challenges of service organisations. Students have the opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. Through practical assignments, students assume full responsibility for planning and executing codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided into groups, and each group will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.) and (2) identify opportunities to apply service codesign methods to create new services. The outcome of this work will be presented in a report that is collaboratively written by team members. Groups will receive feedback from their coaches on an initial draft of the report. This feedback is a formative assessment and does not count towards the final grade for the course. Then, after the delivery of this first report, coaches will establish a grade for assignment 1. This summative assessment represents 30% of a student's final grade. In assignment 2, the groups assembled for the first assignment will plan and execute codesign sessions to address one of the challenges identified before. This work will include a pilot session, which can be done with other students following the course, and another final session with external participants representing the target group. After receiving formative feedback from coaches, as done in the first assignment, groups will again deliver a report with the work done for assignment 2. An additional component of this report is a self-reflection on learning experiences done individually by each student in the group. When evaluating this second report, coaches can add or subtract a maximum of 1,0 points from the grade given the group's work in order to arrive at an individual grade for each student for assignment 2. This grade is summative and will represent 70% of students' final grade for the course.	
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Special Information	Contact: cis-io@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select one

IOB4-P1	Human Factors & Ergonomics	5
Responsible Instructor	Ir. G.P.M. Hoekstra	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications	
Study Goals	This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains. Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	The course will work with the book: Lee, J. D., Wickens, C. D., Liu, Y., & Boyle, L. N. (2017). Designing for people. ShangHai, 173. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. The deliverable for the group assignment (60% of the grade) is a scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report and presentation in week 6.	
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Special Information	Registration beyond the registration deadlines is not possible. Contact: hfe-io@tudelft.nl	

IOB4-P2	Experience, Motivation & Behaviour	5
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective, students will develop a deeper knowledge of human behavior, motivation and experience. A focus will be on understanding how to influence individuals, social and environmental factors as well as tools and strategies to design with. During the course, we will apply the principles of psychology and behavior change on the themes of health, mobility and sustainability. The content of this course is relevant for students, because it will help students to get a better understanding of how design can support people to change their behavior.	
Study Goals	Upon finishing this course students are able to: 1. Explain which factors influence motivation and behavior using frameworks from psychological theories covered in the course (deepening LO2.6.3). 2. Analyze existing behavior change interventions using motivation and behavior frameworks covered in the course. 3. Apply principles of motivation and behavior in their designs to measure, change and evaluate motivation and behavior (deepening LO2.6.3). 4. Apply principles of human experience, emotion and perception in their design. (deepening LO2.6.2) 5. Evaluate design interventions and relate these results to theoretical frameworks that are discussed in the course. 6. Substantiate clearly their insights and design decisions during a presentation and in a written report (deepening LO 6.3.2)	
Education Method	The course consists of two main educational parts: lectures and a group assignments. The lectures will provide students with the information regarding the basic principles of experience, motivation and behavior, and will be performed by different lecturers and guest lecturers. In the assignments, student will apply the content that is provided in the lectures. During the assignments, the students will be supervised by a coach giving direction on their work and deliverables. During the coach meetings, students will present their work and discuss the topic with their peers.	
Literature and Study Materials	A reader will be created with chapters from several books and scientific papers. The book chapters will focus on content, explaining theoretical frameworks and models. Scientific papers will provide examples on how these frameworks are applied in design interventions on different domains and target groups. Recommended literature and study materials: - Kristina Niedderer, Stephen Clune and Geke Ludden (2018) Design for Behaviour change: theories and practices of designing for change. Routledge, NY - Stephen Wendel (2013) Designing for Behavior Change: applying psychology and behavioral economics. O'Reilly Media, Inc.	
Assessment	The learning objectives of the course are assessed through a written report and presentation performed in the last week. The report and presentation provide an overview of the assignment, the insights it created and a critical reflection on their own work. The final grade will be calculated based on the written report (80%) and the presentation (20%).	
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Special Information	Contact: emb-io@tudelft.nl	

IOB4-P3	Culture & Society	5
Responsible Instructor	Dr. A. Singh	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course students will learn how to study cultural and societal processes. The social and cultural perspective will empower students to move beyond the individual as the unit of design. Through close readings of text and design case studies, we will focus on the relationships between people, and between people and their designed environment. The intention of most contemporary designers is to contribute to changes for better futures. This course critically reflects on this concept of change. We will raise and address the questions about what exactly should be changed and what not, for whom these changes are designed, how change happens over time and place, and what the possible roles of the designer in relation to change might be. Insights from this course will inspire the development of students design visions and help them communicate the outcomes of their socio-cultural analysis. We will address four theoretical concepts: Culture About the complexity of defining culture(s), and how culture could be viewed in for designers useful ways. Power About structure and agency: two lenses for studying power, politics and other social phenomena, and how designs mediate power. Identity About the various ways to understand identities, and how design shapes them. Change About the meaning of change and other related concepts such as innovation and transition, and why designers need them ... or not ... Culture & Society A final reflection how insights apply in our daily practices. The content of this course is relevant for students, because what we design shapes cultures and societies and vice versa: their logics and values influence our designs.	
Study Goals	Upon finishing this course students are able: 1. to analyse the anthropological, sociological, political, and economic context that stage and shapes society. (2.8.1/2.8.2) 2. to identify social and cultural issues in design. (3.4.1) 3. to critically reflect on cultural and societal issues. (5.1.1; 5.4.2) 4. to communicate their analysis on culture and society. (6.3.1)	
Education Method	The exchange of thoughts and opinions is essential for critical thinking. Therefore, we choose education methods that enhance the interaction between students, and between students and experts of theoretical concepts and design. The course consists of weekly activities: (interactive) Lectures for the explanation of theory, discussion and time for Q&A; Readings and videos for self-study; Seminars in small groups to present, reflect and discuss key concepts; An assignment resulting in a representation of insights in a portfolio; Intermediate feedback moments (by peers and lecturers). Peer to peer presentation to articulate insights and questions.	
Literature and Study Materials	The course will use a reader with papers, book sections, and links to documentaries.	
Assessment	During the course of the course, partial individual deliverables (as part of the portfolio) are requested, on which formative feedback is given by both peers and lecturers. At the end of the course, the learning objectives of the course are summatively assessed through individual work. The grade will be calculated based on this individual final deliverable.	
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Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Dfi schakelprogramma 2

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Social/Interaction Focus

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren.	
Study Goals	Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek. Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

IOB4-1-24	DP4 Designing for Society	10
Course Coordinator	D. Özkaramanl	
Contact Hours / Week x/x/x/x	4 hours/week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course Design Project 4 (DP4), Designing for Society, is one of the four large design project courses in the bachelor curriculum of Industrial Design Engineering at TU Delft. In this course, we work with social designers and societal stakeholders. This course helps students explore and better understand how design can contribute to addressing social issues in a variety of domains such as food, mobility, mental healthcare, or biodiversity. There is an increasing demand for social designers who are by hired by governments, public and social sector organisations and other industries that want to make a positive impact on society. The course combines gathering deep insights into human needs and experiences, with broad insights into implications for society and the planet. To achieve this, you will learn to use the Social Implication Design methodology (Tromp & Hekkert, 2018) as a framework to structure your design approach. The core of this methodology is the development of a probable, coherent, and original future narrative and a design statement that responds to that narrative by taking a stance on the probable future envisioned. These two main ingredients are used to (re-)frame the challenge you are working on, and they guide the development of concepts. Throughout the course, we will pay particular attention to the role of insights into experience, meaning and behaviour in a design process. The course consists of three assignments, that align with the main stages of the Social Implication Design: Phase one 1 - Group assignment with an individual component. Students start the assignment in groups and by doing desktop research and literature search about their chosen case/challenge. After an initial exploration, they (same as assignment 1 above probing experiment in context) is focused on an exploration of the domain at hand from the perspective of experience and meaning. Finally, they get back into groups to complement insights from assignment 1 with additional context factors (broader societal insights gathered through further research to work towards a future context and a design statement. This helps them to form an intention / stance on how to create change through design. The future context and design statement collectively make up their design vision. Phase 2 - Individual assignment Students develop and visualize a concept that is in line with their design statement; they develop an accompanying design rationale. In addition, they reflect on their role as a future (social) designer in society in an individual positioning essay. Phase 3 - Group assignment Students develop and build an experience prototype of one of the concepts in their group to test the assumptions that were part of their design rationale. They also anticipate and visualize the social implications of their design concept. In addition to learning a new design approach in this course, you will be introduced to the practice of social design, and you will reflect on the role you want to play as a designer in society.	
Study Goals	LO 1: Students are able to deal with ambiguity and uncertainty during the design process in order to make decisions (1.1.4. Ambiguity and uncertainty) LO2: Students are able to systematically explore the context of a given design challenge from different perspectives to build a clear understanding of the design challenge at hand (1.3.1 Explore design challenge) LO3: Students are able to express critical reflection on their own thinking, decision making and actions and adjust these accordingly, and reflect critically towards the domain (5.1.1 Critical thinking) LO4: Students are able to create visualizations, models and prototypes to effectively reflect in action to develop the design (1.5.1 communicating concepts) LO5: Students are able to balance different interests and concerns to develop a design with optimal sustainable impact that is feasible, desirable and viable (1.2.2 Balance interests) LO6: Students are able to identify social issues in design and acknowledge the professional responsibilities of designers (3.4.1 Moral sensitivity) LO7: Students are able to envision a possible future context related to a given situation to formulate a desired state (1.3.2 Envision) LO8: Students are able to identify and prioritise specific elements of the context to reframe a given proposition and formulate realistic goals (1.3.3 Reframe) LO9: Students are able to work effectively in teams (6.2.1 Work in teams) LO10: Students are able to generate concepts that meet the design goal at hand (1.4.2 Generate concepts) LO11: Students are able to generate effective communication means for various purposes (6.3.1 Communication means) LO12: Students are able to make statements about the anticipated sustainable impact of a design outcome on environment, society and economy (1.6.4 Anticipate impact) LO13: Students are able to position themselves in the field of design (5.3.2 Position yourself)	
Education Method	The educational model to learn to design for society is based on a studio-model which includes coaching, activity-based learning and peer learning. Each student and team will have a coach who will be frequently available to provide feedback on your work so far. To get students started on specific steps in the approach, your coach will facilitate workshop-style activities. Finally, you will also help each other by providing formative feedback in a peer learning structure. To guide your design process, you will furthermore have access to tutorial videos and other resources. In addition, we will offer a number of skill-related interactive sessions where students gain hands-on experience with techniques including storyboarding and team-work. In addition to developing a concept in assignment 3, individual students work on a personal position about the role they would like to take as a designer in society. This personal positioning will be supported by peer support groups and a series of activities in a workshop setting where students will reflect on values, worldviews, and strengths. To collectively explore the role of designers in society we will invite practicing social designers to share their experiences with you.	
Literature and Study Materials	Compulsory reading: Tromp, N., & Hekkert, P. (2018). Designing for society: Products and services for a better world. Bloomsbury Publishing.	
Assessment	DP4 includes three summative assessments: Assessment 1: Group component: To develop a future context & design statement for the selected societal challenge (4 weeks). Individual component: To create and apply an experiential probe for a selected societal challenge (4 weeks). A formative assessment will happen mid-way through the assignment.	

Assessment 2: Individual assignment on developing and visualizing a concept based on the future context and design statement (4 -5 weeks), with a formative assessment after about 2 weeks. In addition, this assignment has an individual component which includes the development of a personal positioning essay within the design field and within society.

Assessment 3: Group assignment on developing and testing an experiential prototype and anticipating and visualising individual and social implications of the proposed concept.

Assignment 1: 40%
Assignment 2: 30%
Assignment 3: 30%

Students who score a course grade between 5,0 and 5,9 (with each separate assignment a 5,0 or higher) can complete a compensation assignment to repair the grade at the end of the semester (after assignment 4). The maximum grade for the course with a compensated assignment is 6,0.

Enrolment / Application

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Registration beyond the registration deadlines is not possible.

Special Information

Contact: dp4-io@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select two

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is structured in two pillars. In the first one, students learn through lectures, self-study, and group work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students also have lectures by external guests who provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. The second pillar of the course is about organising codesign sessions to tackle real-world challenges of service organisations. Students have the opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. Through practical assignments, students assume full responsibility for planning and executing codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided into groups, and each group will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.) and (2) identify opportunities to apply service codesign methods to create new services. The outcome of this work will be presented in a report that is collaboratively written by team members. Groups will receive feedback from their coaches on an initial draft of the report. This feedback is a formative assessment and does not count towards the final grade for the course. Then, after the delivery of this first report, coaches will establish a grade for assignment 1. This summative assessment represents 30% of a student's final grade. In assignment 2, the groups assembled for the first assignment will plan and execute codesign sessions to address one of the challenges identified before. This work will include a pilot session, which can be done with other students following the course, and another final session with external participants representing the target group. After receiving formative feedback from coaches, as done in the first assignment, groups will again deliver a report with the work done for assignment 2. An additional component of this report is a self-reflection on learning experiences done individually by each student in the group. When evaluating this second report, coaches can add or subtract a maximum of 1,0 points from the grade given the group's work in order to arrive at an individual grade for each student for assignment 2. This grade is summative and will represent 70% of students' final grade for the course.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cis-io@tudelft.nl	

IOB4-P1	Human Factors & Ergonomics	5
Responsible Instructor	Ir. G.P.M. Hoekstra	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications	
Study Goals	This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains. Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	The course will work with the book: Lee, J. D., Wickens, C. D., Liu, Y., & Boyle, L. N. (2017). Designing for people. ShangHai, 173. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. The deliverable for the group assignment (60% of the grade) is a scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report and presentation in week 6.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: hfe-io@tudelft.nl	

IOB4-P2	Experience, Motivation & Behaviour	5
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective, students will develop a deeper knowledge of human behavior, motivation and experience. A focus will be on understanding how to influence individuals, social and environmental factors as well as tools and strategies to design with. During the course, we will apply the principles of psychology and behavior change on the themes of health, mobility and sustainability. The content of this course is relevant for students, because it will help students to get a better understanding of how design can support people to change their behavior.	
Study Goals	Upon finishing this course students are able to: 1. Explain which factors influence motivation and behavior using frameworks from psychological theories covered in the course (deepening LO2.6.3). 2. Analyze existing behavior change interventions using motivation and behavior frameworks covered in the course. 3. Apply principles of motivation and behavior in their designs to measure, change and evaluate motivation and behavior (deepening LO2.6.3). 4. Apply principles of human experience, emotion and perception in their design. (deepening LO2.6.2) 5. Evaluate design interventions and relate these results to theoretical frameworks that are discussed in the course. 6. Substantiate clearly their insights and design decisions during a presentation and in a written report (deepening LO 6.3.2)	
Education Method	The course consists of two main educational parts: lectures and a group assignments. The lectures will provide students with the information regarding the basic principles of experience, motivation and behavior, and will be performed by different lecturers and guest lecturers. In the assignments, student will apply the content that is provided in the lectures. During the assignments, the students will be supervised by a coach giving direction on their work and deliverables. During the coach meetings, students will present their work and discuss the topic with their peers.	
Literature and Study Materials	A reader will be created with chapters from several books and scientific papers. The book chapters will focus on content, explaining theoretical frameworks and models. Scientific papers will provide examples on how these frameworks are applied in design interventions on different domains and target groups. Recommended literature and study materials: - Kristina Niedderer, Stephen Clune and Geke Ludden (2018) Design for Behaviour change: theories and practices of designing for change. Routledge, NY - Stephen Wendel (2013) Designing for Behavior Change: applying psychology and behavioral economics. O'Reilly Media, Inc.	
Assessment	The learning objectives of the course are assessed through a written report and presentation performed in the last week. The report and presentation provide an overview of the assignment, the insights it created and a critical reflection on their own work. The final grade will be calculated based on the written report (80%) and the presentation (20%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: emb-io@tudelft.nl	

IOB4-P3	Culture & Society	5
Responsible Instructor	Dr. A. Singh	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course students will learn how to study cultural and societal processes. The social and cultural perspective will empower students to move beyond the individual as the unit of design. Through close readings of text and design case studies, we will focus on the relationships between people, and between people and their designed environment. The intention of most contemporary designers is to contribute to changes for better futures. This course critically reflects on this concept of change. We will raise and address the questions about what exactly should be changed and what not, for whom these changes are designed, how change happens over time and place, and what the possible roles of the designer in relation to change might be. Insights from this course will inspire the development of students design visions and help them communicate the outcomes of their socio-cultural analysis. We will address four theoretical concepts: Culture About the complexity of defining culture(s), and how culture could be viewed in for designers useful ways. Power About structure and agency: two lenses for studying power, politics and other social phenomena, and how designs mediate power. Identity About the various ways to understand identities, and how design shapes them. Change About the meaning of change and other related concepts such as innovation and transition, and why designers need them ... or not ... Culture & Society A final reflection how insights apply in our daily practices. The content of this course is relevant for students, because what we design shapes cultures and societies and vice versa: their logics and values influence our designs.	
Study Goals	Upon finishing this course students are able: 1. to analyse the anthropological, sociological, political, and economic context that stage and shapes society. (2.8.1/2.8.2) 2. to identify social and cultural issues in design. (3.4.1) 3. to critically reflect on cultural and societal issues. (5.1.1; 5.4.2) 4. to communicate their analysis on culture and society. (6.3.1)	
Education Method	The exchange of thoughts and opinions is essential for critical thinking. Therefore, we choose education methods that enhance the interaction between students, and between students and experts of theoretical concepts and design. The course consists of weekly activities: (interactive) Lectures for the explanation of theory, discussion and time for Q&A; Readings and videos for self-study; Seminars in small groups to present, reflect and discuss key concepts; An assignment resulting in a representation of insights in a portfolio; Intermediate feedback moments (by peers and lecturers). Peer to peer presentation to articulate insights and questions.	
Literature and Study Materials	The course will use a reader with papers, book sections, and links to documentaries.	
Assessment	During the course of the course, partial individual deliverables (as part of the portfolio) are requested, on which formative feedback is given by both peers and lecturers. At the end of the course, the learning objectives of the course are summatively assessed through individual work. The grade will be calculated based on this individual final deliverable.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cs-io@tudelft.nl	

IOB4-T3	Machine Learning for Design	5
Course Coordinator	Prof.dr.ir. A. Bozzon	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This technology elective will provide students with the knowledge required to understand, design, and evaluate machine learning systems in the context of the design of intelligent products, services, and systems (iPSSs). Machine learning (ML) is a computational approach that aims at giving computers the ability to learn without being explicitly programmed (A. Samuel, 1959). Smart thermostats, voice-based personal assistants, autonomous vehicles, traffic control systems, online social networks, web shopping platforms, content creation platforms, personal health appliances: much of current and future iPSSs are powered by ML technology, influencing and shaping our interests, habits, lives, and society. To meaningfully envision and design future iPSSs that is beneficial and useful to people and society, designers must: engage with the details of how ML systems see the world, reason about it, and interact with it; experience the quirks, biases, and failures of ML technology; contend with how agency, initiative, trust, and explainability mediate the interaction between human and iPSSs; and understand how functionalities enabled by ML can be designed in iPSSs. Students in this course will gain practical experience with ML technology and learn how to think critically of what ML systems can do, and how they could and should be integrated into iPSSs. The course will cover the following topics: 1. AI and ML in intelligent products, services, and systems (design). The students will be introduced to the role that Artificial Intelligence and, more specifically, Machine Learning technology play in iPSSs, in their design process, and in their lifecycle. The most important ML concepts, terminology, paradigms, and methods will also be described and exemplified. 2. Text Processing methods for iPSSs. Building on the knowledge acquired in the DATA course, students will be introduced to existing ML techniques for text processing (parsing, analysis, generation), and will practice with their use in realistic design contexts. 3. Image Processing methods for iPSSs. Students will be introduced to existing ML techniques for image processing (acquisition, analysis, generation), and will practice with their use in realistic design contexts. 4. Creating Machine Learning Models Students. will deepen their knowledge in how machine learning models could be trained, and on how their performance could be evaluated. Specific attention will be devoted to the task of training data a) design, b) creation, and c) usage in realistic adoption scenarios. 5. Designing iPSSs that include Machine Learning technology. In this final module, students will learn how to design (using human-centred approaches) intelligent products, services, and systems. Students will understand which issues could impact the experience of people when interacting with iPSSs, and learn to assess how ML technology can and should be integrated into iPSSs.	
Study Goals	Upon finishing this course students are able to: - LO1: explain common terminology used when discussing about Machine Learning algorithms, systems and their evaluation (deepening LO 2.4.4) - LO2: identify the requirements for a Machine Learning system (deepening LO 2.5.2) - LO3: describe and compare different classes of Machine Learning techniques (deepening LO2.4.1, LO 2.4.4, LO 3.2.3) - LO4: describe and compare design decisions in the context of iPSSs design and defend the relevance and correctness of their choices (deepening LO2.5.1, LO 5.1.1) - LO5: recommend appropriate Machine Learning technique(s) for a certain problem domain (deepening LO1.2.2, LO1.2.3, LO3.4.3, LO 5.2.2) - LO6: assess their team members' work with respect to methodological soundness and contribution. (deepening LO1.5.4, LO 1.6.1, LO 5.1.2)	
Education Method	After a first introductory week, the course will include four modules, each including: (1) two lectures (one per week) where the main theoretical concepts related to the module are introduced and put in context; (2) studio work: during the first week of a module, students will have individual hands-on experience with the technology related to the module, while during the second week students will collaborate in teams to work on their bi-weekly formative assignments (teachers and student assistants are available for supervision and feedback), and (3) self-study based on readings and additional teaching and practical material associated to each module, and related individual formative assignments.	
Literature and Study Materials	Students will be provided with a reader, containing curated readings, lecture notes, tutorials, and Webcasts related to each course module. For individual hands-on experience, students will be provided with examples (templates) of practical assignments. These templates serve as references for the group assignments constituting the assessment elements of the course. Practical work will be performed through interactive web-based resources and online visual programming tools selected to facilitate understanding of ML techniques and models.	
Assessment	The learning objectives of the course are assessed through an individually assessed group work, and an individual exam. Group work (50%): students will be organized in groups of 4 or 5 members. Groups deliver one report for each course module to build a final portfolio of group work that is assessed at the end of the course, and that accounts for 80% of the group work. Students will receive an individual grade for the group work. The individual grade includes an individual assessment - that takes into account input from group members - which accounts for 20% of the group work. Individual exam (50%): a mix of multiple choice and open answer questions. Groups will attend weekly formative coaching sessions. Weekly Individual formative assessment will allow students to test their knowledge of the topics addressed in a bi-weekly module. Multiple choice questions will be automatically evaluated, open answer questions will be evaluated by teaching assistants.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: mlfd-io@tudelft.nl	

IOB6-E8	Design Analytics	5
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period	Dr. H. Verma 4 3 3 3 4	
Course Language Course Contents	English In this technological and methodological elective, students will gather tools and acquire knowledge to critically evaluate (interactive and intelligent) technological prototypes and set the stage for the next iteration of the design phase by eliciting actionable design guidelines based on comprehensive empirical analysis. The technological component of this course is demonstrated by the strong focus on the improvement of IPSS (intelligent products, services, and systems) such as, conversational assistants, wearables, etc. Instead of examining the complete design cycle, students will focus solely on the quantitative assessment of IPSS and examine their cognitive, perceptual, and experiential effects. To this end, students will learn and add skills to their methodological toolkit, which will enable them to a) design sound experiments (with Control and Experimental groups), b) use sensing technologies (e.g., eye-tracking) and adapt existing data-acquisition tools (e.g., NASA TLX, MSLQ), c) identify the right statistical tests (descriptive vs. inferential; comparison vs. regression), d) discriminate correlation from causation and identify confounding factors, and e) understand the proper means of communicating their quantitative analysis and results. The content of this course is relevant for students because they learn how to systematically assess their designs with standardized practices and communicate their findings with multiple stakeholders and a broader design-research community. In this way, this course offers an essential ingredient in students transformation from a UX Designer to a UX Researcher. In this course, the student will become familiar with the 5-stage Design Analytics Process (DAP) i.e., design diagnostics, study design synthesis, experimentation, data analysis, and design prescription.	
Study Goals	Upon finishing this course, students are able to: 1: Describe the design unknowns or flaws of a given product, service, or system (deepening LO 1.3.3, LO 1.6.3). 2: Formulate research questions and pertinent hypotheses to empirically evaluate the given technology (deepening LO 4.2.1). 3: Design an experiment protocol to validate the research hypotheses (deepening LO 4.2.2, LO 1.6.1). 4a: Evaluate the quality of collected multi-modal data (deepening LO 2.4.4). 4b: Apply the pre-designed experiment protocol with real subjects (deepening LO 1.6.1) 5: Analyze experimental data using appropriate statistical methods (deepening LO 2.1.1). 6: Explain the results of statistical analysis for future design improvement (deepening LO 4.3.1, LO 4.3.2).	
Education Method	This 10-week course consists of 8 weekly lectures (2h/week), 8 weekly hands-on workshops (upto 1.5h/week) with Teaching Assistants, and 8 weekly thematic coaching sessions (0.5h/week). Students are expected to prepare themselves for an upcoming lecture by studying the provided reading material and by watching the provided educational videos in a flipped classroom format. Lectures and hands-on-sessions will be related in terms of content to ensure that the theory--in the lectures--will be put into practice--in the hands-on workshops. Moreover, student groups will have weekly discussions with thematic coaches, who will guide them in their particular design briefs. Inherently, this course provides the ideal opportunity for applying an inquiry-based learning approach at scale. As such, we will invite colleagues, interested parties, and stakeholders to submit their IPSS (Intelligent Products, Services, and Systems) designs for evaluation a few months before the start of the course. Group assignments In week 9, the student groups are expected to submit a comprehensive report documenting their engagement with the 5-stage Design Analytics Process, and their empirical evaluation of the chosen IPSS. Weekly sessions with the thematic coach (formative assessments) will allow groups the possibility of seeking feedback on their project direction and progress. Kickstarting lecture and hands-on session (week 1) At the kickstarting lecture, we will employ ice-breaking techniques (e.g., "speed-dating") for enabling students to familiarize among themselves. Next, we will perform a brief diagnostic assessment by using the Kahoot! game-based learning platform. Students will interactively answer questions that determine their personality profiles (e.g., the Big-5), interests, and grouping preferences so that we form effective and diverse student teams (3--4 persons). After a brief presentation of all the available IPSS designs, the newly formed teams will bid for the design they prefer to evaluate throughout this course by delivering a final report at the end of the course. Intermediate lectures and hands-on sessions (weeks 2--8) In each lecture, we will showcase the research and experimental thinking that corresponds to the given stage in the course, and in full alignment with the established LOs. To this end, we will present examples of existing IPSS designs tested through our 5-stage DAP, outlined above, in the form of "evaluation cases": An evaluation case can be a prototype borrowed from literature where we describe how specific design unknowns are targeted for formulating concrete RQs/Hypotheses. During lectures, we will define our threshold concepts--notions that once understood change the perception of a subject, phenomenon, or experience. During the lectures, we will use Kahoot! for increasing engagement with brief quizzes and for collecting learning feedback in situ. In the following hands-on sessions, the comprehension of the threshold concepts will be tested with practical work on other evaluation cases. Then, the students will plan how they will move forward with their evaluation cases in collaboration with the thematic coaches.	
Literature and Study Materials	The course coordinators will design a reader on Brightspace with a curated set of readings, lecture notes, web resources, hands-on tutorials, and videos for different modules within this elective. The mandatory weekly readings will be selected chapters from a list of relevant books, and other articles.	
Assessment	The learning objectives of this elective will be assessed through a group report, including a final presentation, and an individual exam. Group deliverables will contribute to 60% of the students final grade. The final exam will contribute to the remaining 40%. Group Work (60%): Student groups will deliver a final report at the end of the elective (max. 20 pages), accompanied by a final presentation. This group deliverable corresponds to the aforementioned DAP stages and the underlying LOs. The LOs are fulfilled by the rigorous evaluation of an IPSS (design artifact), for which the groups will work throughout the duration of the course. The course coordinators will provide a template and assessment rubric for these reports, and instructions on the length and structure of each report. The final report will manifest into a research portfolio that will be presented publicly at the end of the course to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. The final individual grade for the group work will be the sum of the grades for each group report, weighted by the peer contribution assessment scores produced weekly.	

Enrolment / Application	Individual Exam (40%): A final exam will constitute a mix of MCQs (multiple choice questions) and open questions. Formative Assessment: Weekly thematic coaching will provide students with the necessary feedback as well as enable them to assess the progress of their learning, their conceptual understanding. In addition, thematic coaching will assist students to effectively align their project direction by reflecting critically about the diverse aspects of design analytics. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible
Special Information	Contact: da-io@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Schakelprogramma Industrial Product Design

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

IPD schakelprogramma 1

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Design Focus

IOB2-1-23	DP2: Designing Product Services	10
Course Coordinator	Ir. C.P.J.M. Kroon	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Prerequisites		
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dp2-io@tudelft.nl	

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren. Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek.	
Study Goals	Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is structured in two pillars. In the first one, students learn through lectures, self-study, and group work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students also have lectures by external guests who provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. The second pillar of the course is about organising codesign sessions to tackle real-world challenges of service organisations. Students have the opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. Through practical assignments, students assume full responsibility for planning and executing codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided into groups, and each group will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.) and (2) identify opportunities to apply service codesign methods to create new services. The outcome of this work will be presented in a report that is collaboratively written by team members. Groups will receive feedback from their coaches on an initial draft of the report. This feedback is a formative assessment and does not count towards the final grade for the course. Then, after the delivery of this first report, coaches will establish a grade for assignment 1. This summative assessment represents 30% of a student's final grade. In assignment 2, the groups assembled for the first assignment will plan and execute codesign sessions to address one of the challenges identified before. This work will include a pilot session, which can be done with other students following the course, and another final session with external participants representing the target group. After receiving formative feedback from coaches, as done in the first assignment, groups will again deliver a report with the work done for assignment 2. An additional component of this report is a self-reflection on learning experiences done individually by each student in the group. When evaluating this second report, coaches can add or subtract a maximum of 1,0 points from the grade given the group's work in order to arrive at an individual grade for each student for assignment 2. This grade is summative and will represent 70% of students' final grade for the course.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cis-io@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select one

IOB4-P1	Human Factors & Ergonomics	5
Responsible Instructor	Ir. G.P.M. Hoekstra	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications	
Study Goals	This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains. Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	The course will work with the book: Lee, J. D., Wickens, C. D., Liu, Y., & Boyle, L. N. (2017). Designing for people. ShangHai, 173. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. The deliverable for the group assignment (60% of the grade) is a scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report and presentation in week 6.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: hfe-io@tudelft.nl	

IOB4-P2	Experience, Motivation & Behaviour	5
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective, students will develop a deeper knowledge of human behavior, motivation and experience. A focus will be on understanding how to influence individuals, social and environmental factors as well as tools and strategies to design with. During the course, we will apply the principles of psychology and behavior change on the themes of health, mobility and sustainability. The content of this course is relevant for students, because it will help students to get a better understanding of how design can support people to change their behavior.	
Study Goals	Upon finishing this course students are able to: 1. Explain which factors influence motivation and behavior using frameworks from psychological theories covered in the course (deepening LO2.6.3). 2. Analyze existing behavior change interventions using motivation and behavior frameworks covered in the course. 3. Apply principles of motivation and behavior in their designs to measure, change and evaluate motivation and behavior (deepening LO2.6.3). 4. Apply principles of human experience, emotion and perception in their design. (deepening LO2.6.2) 5. Evaluate design interventions and relate these results to theoretical frameworks that are discussed in the course. 6. Substantiate clearly their insights and design decisions during a presentation and in a written report (deepening LO 6.3.2)	
Education Method	The course consists of two main educational parts: lectures and a group assignments. The lectures will provide students with the information regarding the basic principles of experience, motivation and behavior, and will be performed by different lecturers and guest lecturers. In the assignments, student will apply the content that is provided in the lectures. During the assignments, the students will be supervised by a coach giving direction on their work and deliverables. During the coach meetings, students will present their work and discuss the topic with their peers.	
Literature and Study Materials	A reader will be created with chapters from several books and scientific papers. The book chapters will focus on content, explaining theoretical frameworks and models. Scientific papers will provide examples on how these frameworks are applied in design interventions on different domains and target groups. Recommended literature and study materials: - Kristina Niedderer, Stephen Clune and Geke Ludden (2018) Design for Behaviour change: theories and practices of designing for change. Routledge, NY - Stephen Wendel (2013) Designing for Behavior Change: applying psychology and behavioral economics. O'Reilly Media, Inc.	
Assessment	The learning objectives of the course are assessed through a written report and presentation performed in the last week. The report and presentation provide an overview of the assignment, the insights it created and a critical reflection on their own work. The final grade will be calculated based on the written report (80%) and the presentation (20%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: emb-io@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

IPD schakelprogramma 2

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Technology/Engineering Focus

IOB2-2-24	Digital Interfaces	5
Course Coordinator	Ing. A.J.C. van der Helm	
Education Period	3	
Start Education	3	
Exam Period	3	
	4	
Course Language	Dutch	
Course Contents	Om interactieve producten te ontwerpen moet een ontwerper de basisprincipes van mens-machine interactie kennen. Deze principes worden verzameld uit drie vakgebieden: (1) de psychologie voor het begrijpen of bedenken van de gebruikers ervaring, (2) het ontwerpen voor vormgeving van interface en dynamisch product gedrag en (3) de electronica voor het kennen van componenten en programmeren van een computer om deze componenten te laten samenwerken in de beoogde functie van het product. Het ontwerpen van mens-machine interactie is als het ontwerpen van een theater-stuk met verschillende scenes waarbij de ontwerper vanuit de eerste persoon alle stappen van de gebruiker en de machine moet plannen, uitdenken en in de tijd achter elkaar zetten. Het omhelst zowel de vormgeving van de productvorm als de choreografie van het dynamisch gedrag. Voor kennis over de interactieve technologie wordt voortgebouwd op het vak Understanding Product Engineering. In Digital Interfaces wordt de anatomie van een product-service system behandeld waar het gaat om het meten van gebruiker acties (met sensoren), het schrijven van de software om de metingen om te zetten in een gewenste product reactie (met actuatoren). Voor kennis over het begrijpen en bedenken van de gebruikers ervaring wordt voortgebouwd op het vak Understanding Humans. In Digital Interfaces wordt die kennis verdiept voor toepassing bij mens-machine interactie.	
Study Goals	Als studenten het vak afgerond hebben, zijn ze in staat om: LO1 principes van menselijke ervaring toe te passen, waaronder multi-sensorische esthetische ervaringen, zingeving en emoties om een product-service systeem te ontwikkelen (2.6.2). LO2 principes van vorm, inclusief vormeigenschappen en ervaringsaspecten, toe te passen om een product-service systeem te ontwikkelen (2.7.2). LO3 de principes van dynamische gedragskwaliteiten, fysieke betrokkenheid en zingeving toe te passen om een product-service systeem te ontwikkelen (2.7.1). LO4 kennis over sensorische waarneming toe te passen voor het ontwikkelen van een interactief product-service systeem dat de omgeving en de mens waarneemt en daarop feedback geeft (2.4.2). LO5 kennis van interactieve technologie, inclusief elektronische basiscomponenten, sensoren en actuatoren, en ingebedde systemen toe te passen in de verwezenlijking van een ontwerp (2.3.1).	
Education Method	Het vak maakt gebruik van de volgende leer activiteiten: a) Een plenair college aan het begin van het vak b) Online video colleges met relevante theorie over mens-machine interactie c) Seminars voor het oefenen van de tentamenstof d) Een individueel schriftelijk examen aan het einde van het vak e) Activiteiten voor het maken en ervaren van interactieve prototypen f) Activiteiten voor het evalueren en reflecteren op het ontwerp en de werking van interactive prototypen Studenten werken individueel in het vak. In de eerste week van het vak wordt het mogelijk gemaakt een eigen leerpad te kiezen door het gebied van mens-machine interactie. Elke stap op het leerpad duurt één week en bestaat uit het individueel maken: 1) van een prototype van een interface (embodiment en interactive technologie) en 2) het gezamenlijk evalueren en reflecteren op gemaakte interfaces. Gedurende het vak kan voortschrijdend inzicht gebruikt worden om dit initiële pad aan te passen. Het maken van het technische gedeelte van de interface wordt ondersteund door opvolging van 5 tutorials, waarin de student geleidelijk steeds meer zelf moet inbrengen. Ieder tutorial heeft flexibiliteit in de keuze van sensoren, actuatoren, vormgeving etc. Gedurende de maak activiteiten is ondersteuning aanwezig. De programmeertaal is Python. Deze programmeertaal wordt ook in andere vakken in de bachelor gebruikt, hierdoor is een investering in het leren te werken met Python nuttig. De technische materialen voor het maken van de prototypen is beschikbaar in de Connected Electronics Kit die studenten kunnen kopen op de faculteit. Het evalueren en reflecteren vindt iedere week plaats in peer-feedback sessies met andere studenten van het vak, deze sessies worden gefaciliteerd door coaches. Voor het evalueren en reflecteren maakt Digital Interfaces gebruik van de kennis en routine die ontwikkeld wordt in het vak Research through Design. Door middel van de wekelijkse maak en peer-feedback sessies is een student in staat om mens-machine interactie te ontdekken op een persoonlijke manier, met aandacht voor persoonlijke voorkeuren voor specifieke vormen van mens-machine interactie. Door kennis te nemen van de prototypen van interfaces gemaakt door andere studenten ontstaat een rijk beeld van het vakgebied en de mogelijkheden voor mens-machine interactie. Het vakgebied is dusdanig groot dat het onmogelijk is om het compleet te behandelen. De verwachting is dat de student kennis verdiept in de design projecten en opvolgende verplichte vakken zoals Design Project 2 (iob2-1), Digital Product Development (iob2-2), Product Engineering (iob3-5).	
Literature and Study Materials	De theorie van het vak wordt beschikbaar gesteld op BrightSpace in de vorm van korte video colleges en teksten. De verzameling tutorials voor de maak activiteiten is beschikbaar online en kan later in de bachelor makkelijk gebruikt of gedeeld worden.	
Assessment	Tijdens een peer-feedback sessie worden de leerdoelen beoordeeld op een formatieve manier door middel van feedback van mede studenten. Het behalen van de leerdoelen wordt beoordeeld op basis van een individueel schriftelijk examen aan het einde van het vak en op basis van de maakactiviteiten.	
Enrolment / Application	Het eindcijfer voor het vak bestaat uit 75% examen en 25% maakactiviteiten. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: di-io@tudelft.nl	

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren. Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek.	
Study Goals	Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

IOB2-5-24	Intelligent Products	5
Responsible Instructor	Prof. G.W. Kortuem	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The Intelligent Products course gives students the foundation to design and develop Intelligent products. Intelligent products utilize advanced digital technologies to enhance their functionality, adaptability, or user experience. They integrate connectivity, data and algorithms to enable capabilities such as data collection, analysis, and interaction with users or other devices. Examples of intelligent products include smart thermostats that learn user preferences and adjust temperature settings accordingly, fitness trackers that monitor physical activity and provide feedback for improving health, and voice-activated virtual assistants that respond to natural language commands. This course will use case studies and real-world product examples to motivate and explain basic principles of core digital technologies such as the Internet of Things, Artificial Intelligence, and cloud computing. Students will learn how to prototype intelligent products using Python and ChatGPT.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the basic principles of IoT, AI and cloud computing technologies, and how they contribute to networked intelligent products 2. Explain the working principles of intelligent products and algorithms 3. Prototype and develop intelligent products 4. Analyse technical and human qualities of intelligent products 5. Understand business models relevant for Intelligent Products	
Education Method	The education approach combines classroom-based education, individual self-study and experiential learning. Theoretical knowledge will be introduced through lectures and mandatory readings. Experiential learning is supported by group-based Python software prototyping exercises that foster practical skills and make theoretical knowledge meaningful and relevant. Case studies of intelligent products are used to contextualise theoretical knowledge and practical exercises and establish relevance for design practice.	
Literature and Study Materials	The study material includes lecture notes, readings and tutorials.	
Assessment	Individual Assessment (70%): digital exam Group Assessment (30%): prototyping assignment with a written final report as deliverable.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dpd-io@tudelft.nl	

IOB4-P1	Human Factors & Ergonomics	5
Responsible Instructor	Ir. G.P.M. Hoekstra	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications	
Study Goals	This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains. Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	The course will work with the book: Lee, J. D., Wickens, C. D., Liu, Y., & Boyle, L. N. (2017). Designing for people. ShangHai, 173. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. The deliverable for the group assignment (60% of the grade) is a scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report and presentation in week 6.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: hfe-io@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select one

IOB4-T1-22	Materials and Manufacturing	5
Course Coordinator	Dr. S. Ghodrat	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Materials are of foundational importance for our technological society. Particularly in these times, when we are challenging huge environmental issues like climate change, it is of paramount importance to be aware that use of materials has a crucial environmental impact. In order to reduce our material footprint, we will have to work more and more with recycled materials and one way of doing so is utilising materials that we extract from waste streams. The core of the elective course Materials and Manufacturing (abbreviated as M&M) is designing new products made of bio-waste or bio-based materials such as fruit waste, wood, coffee waste, waste textiles, seaweed, Japanese knotweed and/or any other bio-based materials. E.g. why not use a starch-based or fruit-based material to make an edible lunch box, or design your own bio-based cover for your tablet? Why not use seaweed or Japanese knot-weed mixture to make something for daily utensils as a replacement for unsustainable polymeric ones? What are the ways we can reuse the coffee and tea waste produced daily? How can we transform waste streams into new materials, creating a positive impact and raising awareness in our society? M&M, in quarter 3, aims to grow your awareness in building up such a future. You will explore how to make and combine waste materials to make new bio-based composites as well as how to test and evaluate their technical and non-technical properties towards designing and making a product. You have various choices to define your product; for instance, an (edible) lunch box, tablet / smart device covers, small company PR gift, or you can make a free design by yourself. There are several stakeholders/companies/material developers involved in this course; they are on board to share their journey from materials towards their product(s) and to help you to motivate your material and product of choice. Companies such as FruitLeather Rotterdam, The Green village at TU, Ubuntu (the environmental solutions platform), BlueBlocks, WhyKnot, Bambooder biobased fibers, NPSP biocomposites and many more are involved in this course. You will learn appropriate tools which are needed to design with such new (and often not yet existing) material composites. How can you best model, make and test a product made by a new material designed by yourself? Manufacturing, testing and user aspects of the case studies will be topics in this elective. The content of this course is relevant for you, because you will gain knowledge and insight on how selection of appropriate manufacturing techniques and of materials will condition the design of a product. The performance (technical and experiential) of finished products is ultimately determined by underlying materials and manufacturing characteristics.	
Study Goals	By the end of the course, you should be able to: 1. Categorise the materials based on understanding their common properties and characteristics as well as variety of manufacturing technologies and the related opportunities and limitations for embodiment design (deepening LO2.2.2). 2. Measure and evaluate the mechanical properties of materials with e.g. tensile test, compression test, bending test, hardness, cyclic test (fatigue) etc. In addition, you will learn with the experiential characterization of your newly made materials used in your case studies (deepening LO 2.2.2). 3. Describe and analyse both design possibilities and technological limitations for their selected manufacturing technologies in their embodiment design, including the unique aspects of bio-based materials (deepening LO 2.2.2). 4. Analysing models (SolidWorks + Finite Element Analysis) of more complex shapes and calculating the loaded product and production aspects (deepening LO 2.1.2). 5. Recommending and creating a suitable material and manufacturing method in a given design situation, taking into consideration a given set of boundary conditions and sustainability aspects (deepening LO 3.3.2).	
Education Method	The course consists of lectures, workshops and project assignments. In a series of lectures, you will be taught about bio-based materials in general (wk1+2), SolidWorks with Finite Element simulations Add-on (wk 3), manufacturing (wk 4+5), technical (materials testing) and non-technical/experiential characterization (wk 6+7). During the final week of the course, external stakeholders/partners will deliver lectures. These sessions will focus on the process of moving from a new material concept to the final product. The connecting workshops, planned on Friday afternoon, provide you the opportunity to practice the theory given during the lectures on Wednesday morning. In general, the workshops are intended to be interactive and challenging. In groups of 5 during the weekly 4-hour workshops of which you have some interactive time with your coaches and the rest of the time you will work on your assignment and prepare your deliverables. This gives you the opportunity to develop your physical prototype as well as the 3D model which is evaluated by your test results; while documenting the entire process and findings in a final report (8 page-report with a defined format) and present your result in a short video clip (3-5 min). The culmination will be the final exhibition of your crafted samples and designed productan exciting event where you'll have the opportunity to showcase your project to peers both within and outside the faculty. In order to encourage you to absorb the Wednesday lectures (theories), individual quizzes will be organized at the beginning of each workshop, aiming to assist you in accomplishing the learning objectives.	
Literature and Study Materials	Different chapters from different books will be used as sources and will be addressed by related teachers. All materials (slides) of the lectures in addition, if the teachers referred to a specific document, materials or references in the lectures that are required to understand the slides, you should also read this. Softwares Ansys Granta EduPack (for both materials and manufacturing) SolidWorks Simulation (FEM Add-in) Recommended literature: Internet websites e.g. Wikipedia, video tutorials, related references and papers.	
Assessment	The course learning objectives are evaluated through both group projects and individual exam. 1. Group work (60%) including mini report (20%) (8 page- report) + video clip (20%) (3-5 min)+ final exhibition (20%) 2. Individual exam (40%): Multiple choice questions (on computer) Formative assessment: There will be a quiz at the beginning of each workshop from the topic of the previous lectures. Workshop of week 5 will be your midterm presentation in which you will be presenting the progress of your case studies to your	

Enrolment / Application	peers and receive feedback from your coaches. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.
Special Information	Registration beyond the registration deadlines is not possible. Contact: mm-io@tudelft.nl or Dr. Sepideh Ghodrat (s.ghodrat@tudelft.nl)

IOB4-T2	Mechatronics	5
Course Coordinator	Dr. Y. Song	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Mechatronics course offers unique opportunities for students to design smart, affordable, reliable and versatile consumer products. For this, students will learn knowledge and skills related to electronics, digital signal processing, control, (data-driven) modelling, simulation, prototyping, as well as mechatronic design principles, where the key is to achieve synergy between mechanical and electronic components while striving for added value, e.g. economical and user benefit, in consumer products. Based on prior knowledge on dynamics, electronics and software, and envisioning the future of smart product design, the mechatronic course will further explore the design of mechatronics products by focusing on three key questions: 1. What are the working principles of mechatronic system and how can (micro) electronics be applied in the system? 2. How can Mechatronic design principles be applied in designing mechatronic products and how do these principles affect the product? 3. How can we implement and optimize the performance of mechatronic system by combining the knowledge on physical models and electronics using intelligently balanced control strategies?	
Study Goals	Upon finishing this course, students are able to: LO1: Explain the principles of (micro) electronics for sensors, sensor circuit, digital signal processing, motors, motor control circuit and embedded system (deepening LO 2.3.1); LO2: Design a simple (PID) controlled mechatronic system and select proper components/algorithms for specific applications; LO3: Model and simulate such as a mechatronic system and explain how different parameters influence the behaviour of the system (deepening LO 2.1.2, 2.3.2); LO4: Implement and optimise the system using a (PID) controller to meet the desired needs within realistic boundary conditions; LO5: Assess the implication of mechatronic design principles on the performance of products (deepening LO 2.3.2);	
Education Method	The course consists of lectures, workshops and a project. The Lectures with practical examples (e.g. case studies by external specialists) are held once per week; The workshops (in groups) are held once per week in the first 6 weeks on the topics of the lectures, respectively A project assignment with the focus on mechatronics system.	
Literature and Study Materials	Lecture slides, specified internet resources, The Mechatronics Handbook (Free) will be used as a reference	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report has a maximum of 10 pages (with a video) and will be 50% of the grade. The individual exam will consist of multiple choice and open questions and will be 50% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mechatronics-io@tudelft.nl	

IOB4-T3	Machine Learning for Design	5
Course Coordinator	Prof.dr.ir. A. Bozzon	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This technology elective will provide students with the knowledge required to understand, design, and evaluate machine learning systems in the context of the design of intelligent products, services, and systems (iPSSs). Machine learning (ML) is a computational approach that aims at giving computers the ability to learn without being explicitly programmed (A. Samuel, 1959). Smart thermostats, voice-based personal assistants, autonomous vehicles, traffic control systems, online social networks, web shopping platforms, content creation platforms, personal health appliances: much of current and future iPSSs are powered by ML technology, influencing and shaping our interests, habits, lives, and society. To meaningfully envision and design future iPSSs that is beneficial and useful to people and society, designers must: engage with the details of how ML systems see the world, reason about it, and interact with it; experience the quirks, biases, and failures of ML technology; contend with how agency, initiative, trust, and explainability mediate the interaction between human and iPSSs; and understand how functionalities enabled by ML can be designed in iPSSs. Students in this course will gain practical experience with ML technology and learn how to think critically of what ML systems can do, and how they could and should be integrated into iPSSs. The course will cover the following topics: 1. AI and ML in intelligent products, services, and systems (design). The students will be introduced to the role that Artificial Intelligence and, more specifically, Machine Learning technology play in iPSSs, in their design process, and in their lifecycle. The most important ML concepts, terminology, paradigms, and methods will also be described and exemplified. 2. Text Processing methods for iPSSs. Building on the knowledge acquired in the DATA course, students will be introduced to existing ML techniques for text processing (parsing, analysis, generation), and will practice with their use in realistic design contexts. 3. Image Processing methods for iPSSs. Students will be introduced to existing ML techniques for image processing (acquisition, analysis, generation), and will practice with their use in realistic design contexts. 4. Creating Machine Learning Models Students. will deepen their knowledge in how machine learning models could be trained, and on how their performance could be evaluated. Specific attention will be devoted to the task of training data a) design, b) creation, and c) usage in realistic adoption scenarios. 5. Designing iPSSs that include Machine Learning technology. In this final module, students will learn how to design (using human-centred approaches) intelligent products, services, and systems. Students will understand which issues could impact the experience of people when interacting with iPSSs, and learn to assess how ML technology can and should be integrated into iPSSs.	
Study Goals	Upon finishing this course students are able to: - LO1: explain common terminology used when discussing about Machine Learning algorithms, systems and their evaluation (deepening LO 2.4.4) - LO2: identify the requirements for a Machine Learning system (deepening LO 2.5.2) - LO3: describe and compare different classes of Machine Learning techniques (deepening LO2.4.1, LO 2.4.4, LO 3.2.3) - LO4: describe and compare design decisions in the context of iPSSs design and defend the relevance and correctness of their choices (deepening LO2.5.1, LO 5.1.1) - LO5: recommend appropriate Machine Learning technique(s) for a certain problem domain (deepening LO1.2.2, LO1.2.3, LO3.4.3, LO 5.2.2) - LO6: assess their team members' work with respect to methodological soundness and contribution. (deepening LO1.5.4, LO 1.6.1, LO 5.1.2)	
Education Method	After a first introductory week, the course will include four modules, each including: (1) two lectures (one per week) where the main theoretical concepts related to the module are introduced and put in context; (2) studio work: during the first week of a module, students will have individual hands-on experience with the technology related to the module, while during the second week students will collaborate in teams to work on their bi-weekly formative assignments (teachers and student assistants are available for supervision and feedback), and (3) self-study based on readings and additional teaching and practical material associated to each module, and related individual formative assignments.	
Literature and Study Materials	Students will be provided with a reader, containing curated readings, lecture notes, tutorials, and Webcasts related to each course module. For individual hands-on experience, students will be provided with examples (templates) of practical assignments. These templates serve as references for the group assignments constituting the assessment elements of the course. Practical work will be performed through interactive web-based resources and online visual programming tools selected to facilitate understanding of ML techniques and models.	
Assessment	The learning objectives of the course are assessed through an individually assessed group work, and an individual exam. Group work (50%): students will be organized in groups of 4 or 5 members. Groups deliver one report for each course module to build a final portfolio of group work that is assessed at the end of the course, and that accounts for 80% of the group work. Students will receive an individual grade for the group work. The individual grade includes an individual assessment - that takes into account input from group members - which accounts for 20% of the group work. Individual exam (50%): a mix of multiple choice and open answer questions. Groups will attend weekly formative coaching sessions. Weekly Individual formative assessment will allow students to test their knowledge of the topics addressed in a bi-weekly module. Multiple choice questions will be automatically evaluated, open answer questions will be evaluated by teaching assistants.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: mlfd-io@tudelft.nl	

IOB6-E8	Design Analytics	5
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period	Dr. H. Verma 4 3 3 3 4	
Course Language Course Contents	English In this technological and methodological elective, students will gather tools and acquire knowledge to critically evaluate (interactive and intelligent) technological prototypes and set the stage for the next iteration of the design phase by eliciting actionable design guidelines based on comprehensive empirical analysis. The technological component of this course is demonstrated by the strong focus on the improvement of IPSS (intelligent products, services, and systems) such as, conversational assistants, wearables, etc. Instead of examining the complete design cycle, students will focus solely on the quantitative assessment of IPSS and examine their cognitive, perceptual, and experiential effects. To this end, students will learn and add skills to their methodological toolkit, which will enable them to a) design sound experiments (with Control and Experimental groups), b) use sensing technologies (e.g., eye-tracking) and adapt existing data-acquisition tools (e.g., NASA TLX, MSLQ), c) identify the right statistical tests (descriptive vs. inferential; comparison vs. regression), d) discriminate correlation from causation and identify confounding factors, and e) understand the proper means of communicating their quantitative analysis and results. The content of this course is relevant for students because they learn how to systematically assess their designs with standardized practices and communicate their findings with multiple stakeholders and a broader design-research community. In this way, this course offers an essential ingredient in students transformation from a UX Designer to a UX Researcher. In this course, the student will become familiar with the 5-stage Design Analytics Process (DAP) i.e., design diagnostics, study design synthesis, experimentation, data analysis, and design prescription.	
Study Goals	Upon finishing this course, students are able to: 1: Describe the design unknowns or flaws of a given product, service, or system (deepening LO 1.3.3, LO 1.6.3). 2: Formulate research questions and pertinent hypotheses to empirically evaluate the given technology (deepening LO 4.2.1). 3: Design an experiment protocol to validate the research hypotheses (deepening LO 4.2.2, LO 1.6.1). 4a: Evaluate the quality of collected multi-modal data (deepening LO 2.4.4). 4b: Apply the pre-designed experiment protocol with real subjects (deepening LO 1.6.1) 5: Analyze experimental data using appropriate statistical methods (deepening LO 2.1.1). 6: Explain the results of statistical analysis for future design improvement (deepening LO 4.3.1, LO 4.3.2).	
Education Method	This 10-week course consists of 8 weekly lectures (2h/week), 8 weekly hands-on workshops (upto 1.5h/week) with Teaching Assistants, and 8 weekly thematic coaching sessions (0.5h/week). Students are expected to prepare themselves for an upcoming lecture by studying the provided reading material and by watching the provided educational videos in a flipped classroom format. Lectures and hands-on-sessions will be related in terms of content to ensure that the theory--in the lectures--will be put into practice--in the hands-on workshops. Moreover, student groups will have weekly discussions with thematic coaches, who will guide them in their particular design briefs. Inherently, this course provides the ideal opportunity for applying an inquiry-based learning approach at scale. As such, we will invite colleagues, interested parties, and stakeholders to submit their IPSS (Intelligent Products, Services, and Systems) designs for evaluation a few months before the start of the course. Group assignments In week 9, the student groups are expected to submit a comprehensive report documenting their engagement with the 5-stage Design Analytics Process, and their empirical evaluation of the chosen IPSS. Weekly sessions with the thematic coach (formative assessments) will allow groups the possibility of seeking feedback on their project direction and progress. Kickstarting lecture and hands-on session (week 1) At the kickstarting lecture, we will employ ice-breaking techniques (e.g., "speed-dating") for enabling students to familiarize among themselves. Next, we will perform a brief diagnostic assessment by using the Kahoot! game-based learning platform. Students will interactively answer questions that determine their personality profiles (e.g., the Big-5), interests, and grouping preferences so that we form effective and diverse student teams (3--4 persons). After a brief presentation of all the available IPSS designs, the newly formed teams will bid for the design they prefer to evaluate throughout this course by delivering a final report at the end of the course. Intermediate lectures and hands-on sessions (weeks 2--8) In each lecture, we will showcase the research and experimental thinking that corresponds to the given stage in the course, and in full alignment with the established LOs. To this end, we will present examples of existing IPSS designs tested through our 5-stage DAP, outlined above, in the form of "evaluation cases": An evaluation case can be a prototype borrowed from literature where we describe how specific design unknowns are targeted for formulating concrete RQs/Hypotheses. During lectures, we will define our threshold concepts--notions that once understood change the perception of a subject, phenomenon, or experience. During the lectures, we will use Kahoot! for increasing engagement with brief quizzes and for collecting learning feedback in situ. In the following hands-on sessions, the comprehension of the threshold concepts will be tested with practical work on other evaluation cases. Then, the students will plan how they will move forward with their evaluation cases in collaboration with the thematic coaches.	
Literature and Study Materials	The course coordinators will design a reader on Brightspace with a curated set of readings, lecture notes, web resources, hands-on tutorials, and videos for different modules within this elective. The mandatory weekly readings will be selected chapters from a list of relevant books, and other articles.	
Assessment	The learning objectives of this elective will be assessed through a group report, including a final presentation, and an individual exam. Group deliverables will contribute to 60% of the students final grade. The final exam will contribute to the remaining 40%. Group Work (60%): Student groups will deliver a final report at the end of the elective (max. 20 pages), accompanied by a final presentation. This group deliverable corresponds to the aforementioned DAP stages and the underlying LOs. The LOs are fulfilled by the rigorous evaluation of an IPSS (design artifact), for which the groups will work throughout the duration of the course. The course coordinators will provide a template and assessment rubric for these reports, and instructions on the length and structure of each report. The final report will manifest into a research portfolio that will be presented publicly at the end of the course to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. The final individual grade for the group work will be the sum of the grades for each group report, weighted by the peer contribution assessment scores produced weekly.	

Enrolment / Application	Individual Exam (40%): A final exam will constitute a mix of MCQs (multiple choice questions) and open questions. Formative Assessment: Weekly thematic coaching will provide students with the necessary feedback as well as enable them to assess the progress of their learning, their conceptual understanding. In addition, thematic coaching will assist students to effectively align their project direction by reflecting critically about the diverse aspects of design analytics. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible
Special Information	Contact: da-io@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Schakelprogramma Strategic Product Design

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

SPD schakelprogramma 1

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Design Focus

IOB2-1-23	DP2: Designing Product Services	10
Course Coordinator	Ir. C.P.J.M. Kroon	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Prerequisites		
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dp2-io@tudelft.nl	

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren.	
Study Goals	Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek. Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen.	
Enrolment / Application	Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%). For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select one

IOB4-B1-22	Strategic Brand Management	5
Responsible Instructor	Ir. S. Wu	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Brands define and differentiate organisations, drive performance and even shape the marketplace. In this elective students will be introduced to the fundamental concepts in brands and branding. They will find out how brand managers cultivate a brand position that resonates with consumers and sets them apart from competition, how brands can inform innovation and how design brings value to the process of branding. They will learn about the strategic and tactical decisions brand managers face in building and managing brands to maintain relevance and fuel growth.	
Study Goals	Upon finishing this course students are able to: LO1. Determine and motivate the positioning of a new brand in relation to opportunities in the market (extending LO2.9.1). LO2. Select, motivate, and develop brand touchpoints to realise the desired brand positioning for a new brand (extending LO2.10.3). LO3. Appraise possible management and innovation options for the brand to propose, motivate and elaborate a meaningful way to fuel brand relevance and growth (extending LO2.9.4).	
Education Method	The course consist of weekly lectures and tutorials, and a project assignment on which students will work as a group (around 4) for the duration of the course. Lectures introduce the theoretical material. Tutorials are a mix of studio discussions and exercises (to promote deeper understanding of the theoretical material and illustrate real world applications of theoretical content), and group feedback meetings and presentations (see Assessment). The brand building project entails the application of course material in the context of a new brand.	
Literature and Study Materials	Books or book chapters, academic/practitioner articles, online learning sources, reports, case texts. Full list of readings and access information will be available on Brightspace.	
Assessment	The learning objectives of the course are assessed through the brand building project report (group; due end of course). The report grade will be adjusted (up or down) for individual contribution based on peer assessment of group members contribution to the process and end result. Students will receive formative feedback on their brand building assignment at the group presentations and feedback sessions. Midway through the course, groups will be offered the opportunity to submit a draft report, should they wish to receive more targeted feedback.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: sbm-io@tudelft.nl	

IOB4-B2	Business Model Design	5
Course Coordinator	Dr. M. Bos-de Vos	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The Business Model Design elective is a course on business models specifically developed for design students. It provides bachelor students with an understanding of what business models are, how they can be designed and what the challenges and opportunities for business model design are. Starting with business models at the firm level (Part 1), the course will gradually move to business model design from the perspective of an interorganizational collaboration (Part 2), and an entire ecosystem (Part 3). Through working on a real-life business model design challenge of a case company from practice, students will acquire new knowledge, methods and skills that allow them to meaningfully engage in business model design activities in diverse situations, including new ventures, creative studios and networks of organizations. The business model design knowledge, methods and skills will be instrumental for the projects that students will execute in following courses (such as design projects and master courses in strategic design) and for determining the role they wish to have as a (strategic) design professional.	
Study Goals	Upon finishing this course students are able to: 1. Recognize and explain BM definitions and components. 2. Distinguish between different business model archetypes (deepening LO2.10.2) 3. Construct clear arguments for choosing a particular business model design and business model design method. 4. Identify relevant components for a new business model design, and map the value flows to show what the value for each stakeholder is (deepening LO 2.10.1). 5. Construct a business model design for a firm, an interorganizational collaboration, and ecosystem using existing methods. 6. Reflect on critical aspects of business model designs and formulate a statement that addresses or solves a business model design challenge in a particular context.	
Education Method	The course consists of a blended learning approach in which online materials, literature discussion groups, and business model design workshops are combined. The online materials (course information, literature, videos) are provided at the start of the course and can be studied in a self-paced manner. The literature discussion groups are held every two weeks (week 3.2, 3.4, 3.6). On Tuesdays, students discuss and work with the literature via an assignment that is provided in the studio that day. This serves as preparation for an in-depth plenary discussion with an expert and a business model design practitioner on Thursdays. The assignments and literature discussion groups allow students to work on study goals 1-3 and to prepare for the exam. Students need to come well-prepared on both days to have meaningful interactions. The business model design workshops are held in the other weeks (week 3.3, 3.5, 3.7) and focus on an existing business model design challenge of a case company from practice. During each of the three workshops, students will work on the successive parts of their assignment. Students will start the workshop on Tuesdays within their group. The instructions for the workshop are provided online and in the studio that day. On Thursdays, the workshops will continue and an expert and a representative of the case company will be available to provide formative feedback on the designs that are made. Afterwards, students will reflect on the results of their workshop. The workshops allow students to bring into practice what they have learned from the literature discussions, while working on study goals 4-6. They also help students to complete the full design assignment. All students have the opportunity to present or debate a selection of Business Model Designs from the assignment with the case company at the end of the course in a dragons den format.	
Literature and Study Materials	Mandatory literature & study materials: - Academic book chapters and articles (per topic). - Videos and materials on business model design methodology. For the literature discussions, students can study one of the provided documents per week, which is mandatory. Through sharing summaries of the articles within their group, working on the assignments with the group, and actively participating in the plenary discussion sessions with experts, students will also learn about the other documents. The exam questions are focused on specific business model design topics and can be answered by having read different sources.	
Assessment	The learning objectives of the course are assessed through an individual exam and an assignment. The individual exam will consist of multiple choice and open questions and will be 50% of the grade. The assignment consists of four parts: the three business model designs that were developed during the workshops including argumentation (these parts will be delivered with a group), and a business model design statement. The fourth part of the assignment can be developed individually or in a group with maximum 3 students. The full assignment accounts for 50% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: bmd-io@tudelft.nl	

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is structured in two pillars. In the first one, students learn through lectures, self-study, and group work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students also have lectures by external guests who provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. The second pillar of the course is about organising codesign sessions to tackle real-world challenges of service organisations. Students have the opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. Through practical assignments, students assume full responsibility for planning and executing codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided into groups, and each group will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.) and (2) identify opportunities to apply service codesign methods to create new services. The outcome of this work will be presented in a report that is collaboratively written by team members. Groups will receive feedback from their coaches on an initial draft of the report. This feedback is a formative assessment and does not count towards the final grade for the course. Then, after the delivery of this first report, coaches will establish a grade for assignment 1. This summative assessment represents 30% of a student's final grade. In assignment 2, the groups assembled for the first assignment will plan and execute codesign sessions to address one of the challenges identified before. This work will include a pilot session, which can be done with other students following the course, and another final session with external participants representing the target group. After receiving formative feedback from coaches, as done in the first assignment, groups will again deliver a report with the work done for assignment 2. An additional component of this report is a self-reflection on learning experiences done individually by each student in the group. When evaluating this second report, coaches can add or subtract a maximum of 1,0 points from the grade given the group's work in order to arrive at an individual grade for each student for assignment 2. This grade is summative and will represent 70% of students' final grade for the course.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cis-io@tudelft.nl	

IOB4-P1	Human Factors & Ergonomics	5
Responsible Instructor	Ir. G.P.M. Hoekstra	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications	
Study Goals	This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains. Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	The course will work with the book: Lee, J. D., Wickens, C. D., Liu, Y., & Boyle, L. N. (2017). Designing for people. ShangHai, 173. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. The deliverable for the group assignment (60% of the grade) is a scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report and presentation in week 6.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: hfe-io@tudelft.nl	

IOB4-P2	Experience, Motivation & Behaviour	5
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective, students will develop a deeper knowledge of human behavior, motivation and experience. A focus will be on understanding how to influence individuals, social and environmental factors as well as tools and strategies to design with. During the course, we will apply the principles of psychology and behavior change on the themes of health, mobility and sustainability. The content of this course is relevant for students, because it will help students to get a better understanding of how design can support people to change their behavior.	
Study Goals	Upon finishing this course students are able to: 1. Explain which factors influence motivation and behavior using frameworks from psychological theories covered in the course (deepening LO2.6.3). 2. Analyze existing behavior change interventions using motivation and behavior frameworks covered in the course. 3. Apply principles of motivation and behavior in their designs to measure, change and evaluate motivation and behavior (deepening LO2.6.3). 4. Apply principles of human experience, emotion and perception in their design. (deepening LO2.6.2) 5. Evaluate design interventions and relate these results to theoretical frameworks that are discussed in the course. 6. Substantiate clearly their insights and design decisions during a presentation and in a written report (deepening LO 6.3.2)	
Education Method	The course consists of two main educational parts: lectures and a group assignments. The lectures will provide students with the information regarding the basic principles of experience, motivation and behavior, and will be performed by different lecturers and guest lecturers. In the assignments, student will apply the content that is provided in the lectures. During the assignments, the students will be supervised by a coach giving direction on their work and deliverables. During the coach meetings, students will present their work and discuss the topic with their peers.	
Literature and Study Materials	A reader will be created with chapters from several books and scientific papers. The book chapters will focus on content, explaining theoretical frameworks and models. Scientific papers will provide examples on how these frameworks are applied in design interventions on different domains and target groups. Recommended literature and study materials: - Kristina Niedderer, Stephen Clune and Geke Ludden (2018) Design for Behaviour change: theories and practices of designing for change. Routledge, NY - Stephen Wendel (2013) Designing for Behavior Change: applying psychology and behavioral economics. O'Reilly Media, Inc.	
Assessment	The learning objectives of the course are assessed through a written report and presentation performed in the last week. The report and presentation provide an overview of the assignment, the insights it created and a critical reflection on their own work. The final grade will be calculated based on the written report (80%) and the presentation (20%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: emb-io@tudelft.nl	

IOB4-P3	Culture & Society	5
Responsible Instructor	Dr. A. Singh	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course students will learn how to study cultural and societal processes. The social and cultural perspective will empower students to move beyond the individual as the unit of design. Through close readings of text and design case studies, we will focus on the relationships between people, and between people and their designed environment. The intention of most contemporary designers is to contribute to changes for better futures. This course critically reflects on this concept of change. We will raise and address the questions about what exactly should be changed and what not, for whom these changes are designed, how change happens over time and place, and what the possible roles of the designer in relation to change might be. Insights from this course will inspire the development of students design visions and help them communicate the outcomes of their socio-cultural analysis. We will address four theoretical concepts: Culture About the complexity of defining culture(s), and how culture could be viewed in for designers useful ways. Power About structure and agency: two lenses for studying power, politics and other social phenomena, and how designs mediate power. Identity About the various ways to understand identities, and how design shapes them. Change About the meaning of change and other related concepts such as innovation and transition, and why designers need them ... or not ... Culture & Society A final reflection how insights apply in our daily practices. The content of this course is relevant for students, because what we design shapes cultures and societies and vice versa: their logics and values influence our designs.	
Study Goals	Upon finishing this course students are able: 1. to analyse the anthropological, sociological, political, and economic context that stage and shapes society. (2.8.1/2.8.2) 2. to identify social and cultural issues in design. (3.4.1) 3. to critically reflect on cultural and societal issues. (5.1.1; 5.4.2) 4. to communicate their analysis on culture and society. (6.3.1)	
Education Method	The exchange of thoughts and opinions is essential for critical thinking. Therefore, we choose education methods that enhance the interaction between students, and between students and experts of theoretical concepts and design. The course consists of weekly activities: (interactive) Lectures for the explanation of theory, discussion and time for Q&A; Readings and videos for self-study; Seminars in small groups to present, reflect and discuss key concepts; An assignment resulting in a representation of insights in a portfolio; Intermediate feedback moments (by peers and lecturers). Peer to peer presentation to articulate insights and questions.	
Literature and Study Materials	The course will use a reader with papers, book sections, and links to documentaries.	
Assessment	During the course of the course, partial individual deliverables (as part of the portfolio) are requested, on which formative feedback is given by both peers and lecturers. At the end of the course, the learning objectives of the course are summatively assessed through individual work. The grade will be calculated based on this individual final deliverable.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cs-io@tudelft.nl	

IOB6-E8	Design Analytics	5
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period	Dr. H. Verma 4 3 3 3 4	
Course Language Course Contents	English In this technological and methodological elective, students will gather tools and acquire knowledge to critically evaluate (interactive and intelligent) technological prototypes and set the stage for the next iteration of the design phase by eliciting actionable design guidelines based on comprehensive empirical analysis. The technological component of this course is demonstrated by the strong focus on the improvement of IPSS (intelligent products, services, and systems) such as, conversational assistants, wearables, etc. Instead of examining the complete design cycle, students will focus solely on the quantitative assessment of IPSS and examine their cognitive, perceptual, and experiential effects. To this end, students will learn and add skills to their methodological toolkit, which will enable them to a) design sound experiments (with Control and Experimental groups), b) use sensing technologies (e.g., eye-tracking) and adapt existing data-acquisition tools (e.g., NASA TLX, MSLQ), c) identify the right statistical tests (descriptive vs. inferential; comparison vs. regression), d) discriminate correlation from causation and identify confounding factors, and e) understand the proper means of communicating their quantitative analysis and results. The content of this course is relevant for students because they learn how to systematically assess their designs with standardized practices and communicate their findings with multiple stakeholders and a broader design-research community. In this way, this course offers an essential ingredient in students transformation from a UX Designer to a UX Researcher. In this course, the student will become familiar with the 5-stage Design Analytics Process (DAP) i.e., design diagnostics, study design synthesis, experimentation, data analysis, and design prescription.	
Study Goals	Upon finishing this course, students are able to: 1: Describe the design unknowns or flaws of a given product, service, or system (deepening LO 1.3.3, LO 1.6.3). 2: Formulate research questions and pertinent hypotheses to empirically evaluate the given technology (deepening LO 4.2.1). 3: Design an experiment protocol to validate the research hypotheses (deepening LO 4.2.2, LO 1.6.1). 4a: Evaluate the quality of collected multi-modal data (deepening LO 2.4.4). 4b: Apply the pre-designed experiment protocol with real subjects (deepening LO 1.6.1) 5: Analyze experimental data using appropriate statistical methods (deepening LO 2.1.1). 6: Explain the results of statistical analysis for future design improvement (deepening LO 4.3.1, LO 4.3.2).	
Education Method	This 10-week course consists of 8 weekly lectures (2h/week), 8 weekly hands-on workshops (upto 1.5h/week) with Teaching Assistants, and 8 weekly thematic coaching sessions (0.5h/week). Students are expected to prepare themselves for an upcoming lecture by studying the provided reading material and by watching the provided educational videos in a flipped classroom format. Lectures and hands-on-sessions will be related in terms of content to ensure that the theory--in the lectures--will be put into practice--in the hands-on workshops. Moreover, student groups will have weekly discussions with thematic coaches, who will guide them in their particular design briefs. Inherently, this course provides the ideal opportunity for applying an inquiry-based learning approach at scale. As such, we will invite colleagues, interested parties, and stakeholders to submit their IPSS (Intelligent Products, Services, and Systems) designs for evaluation a few months before the start of the course. Group assignments In week 9, the student groups are expected to submit a comprehensive report documenting their engagement with the 5-stage Design Analytics Process, and their empirical evaluation of the chosen IPSS. Weekly sessions with the thematic coach (formative assessments) will allow groups the possibility of seeking feedback on their project direction and progress. Kickstarting lecture and hands-on session (week 1) At the kickstarting lecture, we will employ ice-breaking techniques (e.g., "speed-dating") for enabling students to familiarize among themselves. Next, we will perform a brief diagnostic assessment by using the Kahoot! game-based learning platform. Students will interactively answer questions that determine their personality profiles (e.g., the Big-5), interests, and grouping preferences so that we form effective and diverse student teams (3--4 persons). After a brief presentation of all the available IPSS designs, the newly formed teams will bid for the design they prefer to evaluate throughout this course by delivering a final report at the end of the course. Intermediate lectures and hands-on sessions (weeks 2--8) In each lecture, we will showcase the research and experimental thinking that corresponds to the given stage in the course, and in full alignment with the established LOs. To this end, we will present examples of existing IPSS designs tested through our 5-stage DAP, outlined above, in the form of "evaluation cases": An evaluation case can be a prototype borrowed from literature where we describe how specific design unknowns are targeted for formulating concrete RQs/Hypotheses. During lectures, we will define our threshold concepts--notions that once understood change the perception of a subject, phenomenon, or experience. During the lectures, we will use Kahoot! for increasing engagement with brief quizzes and for collecting learning feedback in situ. In the following hands-on sessions, the comprehension of the threshold concepts will be tested with practical work on other evaluation cases. Then, the students will plan how they will move forward with their evaluation cases in collaboration with the thematic coaches.	
Literature and Study Materials	The course coordinators will design a reader on Brightspace with a curated set of readings, lecture notes, web resources, hands-on tutorials, and videos for different modules within this elective. The mandatory weekly readings will be selected chapters from a list of relevant books, and other articles.	
Assessment	The learning objectives of this elective will be assessed through a group report, including a final presentation, and an individual exam. Group deliverables will contribute to 60% of the students final grade. The final exam will contribute to the remaining 40%. Group Work (60%): Student groups will deliver a final report at the end of the elective (max. 20 pages), accompanied by a final presentation. This group deliverable corresponds to the aforementioned DAP stages and the underlying LOs. The LOs are fulfilled by the rigorous evaluation of an IPSS (design artifact), for which the groups will work throughout the duration of the course. The course coordinators will provide a template and assessment rubric for these reports, and instructions on the length and structure of each report. The final report will manifest into a research portfolio that will be presented publicly at the end of the course to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. The final individual grade for the group work will be the sum of the grades for each group report, weighted by the peer contribution assessment scores produced weekly.	

Enrolment / Application	Individual Exam (40%): A final exam will constitute a mix of MCQs (multiple choice questions) and open questions. Formative Assessment: Weekly thematic coaching will provide students with the necessary feedback as well as enable them to assess the progress of their learning, their conceptual understanding. In addition, thematic coaching will assist students to effectively align their project direction by reflecting critically about the diverse aspects of design analytics. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible
Special Information	Contact: da-io@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

SPD schakelprogramma 2

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Organizational Focus

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren. Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek.	
Study Goals	Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerpoplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Responsible Instructor	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course, students are able to: 1. apply principles of human motivation and consumer behaviour related to values in their designs 2. distinguish between different types of business models 3. set up a value proposition of a new design for an organisation 4. identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is 5. identify social issues in design and acknowledge the professional responsibilities of designers 6. analyse the ethical values of a design challenge and use these as inspiration for concept development 7. morally justify design decisions and discuss ethical issues in different settings	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures on Monday morning provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions (on Wednesday afternoon) give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice/short answer questions and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

IOB4-1-24	DP4 Designing for Society	10
Course Coordinator	D. Özkaramanl	
Contact Hours / Week x/x/x/x	4 hours/week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course Design Project 4 (DP4), Designing for Society, is one of the four large design project courses in the bachelor curriculum of Industrial Design Engineering at TU Delft. In this course, we work with social designers and societal stakeholders. This course helps students explore and better understand how design can contribute to addressing social issues in a variety of domains such as food, mobility, mental healthcare, or biodiversity. There is an increasing demand for social designers who are by hired by governments, public and social sector organisations and other industries that want to make a positive impact on society. The course combines gathering deep insights into human needs and experiences, with broad insights into implications for society and the planet. To achieve this, you will learn to use the Social Implication Design methodology (Tromp & Hekkert, 2018) as a framework to structure your design approach. The core of this methodology is the development of a probable, coherent, and original future narrative and a design statement that responds to that narrative by taking a stance on the probable future envisioned. These two main ingredients are used to (re-)frame the challenge you are working on, and they guide the development of concepts. Throughout the course, we will pay particular attention to the role of insights into experience, meaning and behaviour in a design process. The course consists of three assignments, that align with the main stages of the Social Implication Design: Phase one 1 - Group assignment with an individual component. Students start the assignment in groups and by doing desktop research and literature search about their chosen case/challenge. After an initial exploration, they (same as assignment 1 above probing experiment in context) is focused on an exploration of the domain at hand from the perspective of experience and meaning. Finally, they get back into groups to complement insights from assignment 1 with additional context factors (broader societal insights gathered through further research to work towards a future context and a design statement. This helps them to form an intention / stance on how to create change through design. The future context and design statement collectively make up their design vision. Phase 2 - Individual assignment Students develop and visualize a concept that is in line with their design statement; they develop an accompanying design rationale. In addition, they reflect on their role as a future (social) designer in society in an individual positioning essay. Phase 3 - Group assignment Students develop and build an experience prototype of one of the concepts in their group to test the assumptions that were part of their design rationale. They also anticipate and visualize the social implications of their design concept. In addition to learning a new design approach in this course, you will be introduced to the practice of social design, and you will reflect on the role you want to play as a designer in society.	
Study Goals	LO 1: Students are able to deal with ambiguity and uncertainty during the design process in order to make decisions (1.1.4. Ambiguity and uncertainty) LO2: Students are able to systematically explore the context of a given design challenge from different perspectives to build a clear understanding of the design challenge at hand (1.3.1 Explore design challenge) LO3: Students are able to express critical reflection on their own thinking, decision making and actions and adjust these accordingly, and reflect critically towards the domain (5.1.1 Critical thinking) LO4: Students are able to create visualizations, models and prototypes to effectively reflect in action to develop the design (1.5.1 communicating concepts) LO5: Students are able to balance different interests and concerns to develop a design with optimal sustainable impact that is feasible, desirable and viable (1.2.2 Balance interests) LO6: Students are able to identify social issues in design and acknowledge the professional responsibilities of designers (3.4.1 Moral sensitivity) LO7: Students are able to envision a possible future context related to a given situation to formulate a desired state (1.3.2 Envision) LO8: Students are able to identify and prioritise specific elements of the context to reframe a given proposition and formulate realistic goals (1.3.3 Reframe) LO9: Students are able to work effectively in teams (6.2.1 Work in teams) LO10: Students are able to generate concepts that meet the design goal at hand (1.4.2 Generate concepts) LO11: Students are able to generate effective communication means for various purposes (6.3.1 Communication means) LO12: Students are able to make statements about the anticipated sustainable impact of a design outcome on environment, society and economy (1.6.4 Anticipate impact) LO13: Students are able to position themselves in the field of design (5.3.2 Position yourself)	
Education Method	The educational model to learn to design for society is based on a studio-model which includes coaching, activity-based learning and peer learning. Each student and team will have a coach who will be frequently available to provide feedback on your work so far. To get students started on specific steps in the approach, your coach will facilitate workshop-style activities. Finally, you will also help each other by providing formative feedback in a peer learning structure. To guide your design process, you will furthermore have access to tutorial videos and other resources. In addition, we will offer a number of skill-related interactive sessions where students gain hands-on experience with techniques including storyboarding and team-work. In addition to developing a concept in assignment 3, individual students work on a personal position about the role they would like to take as a designer in society. This personal positioning will be supported by peer support groups and a series of activities in a workshop setting where students will reflect on values, worldviews, and strengths. To collectively explore the role of designers in society we will invite practicing social designers to share their experiences with you.	
Literature and Study Materials	Compulsory reading: Tromp, N., & Hekkert, P. (2018). Designing for society: Products and services for a better world. Bloomsbury Publishing.	
Assessment	DP4 includes three summative assessments: Assessment 1: Group component: To develop a future context & design statement for the selected societal challenge (4 weeks). Individual component: To create and apply an experiential probe for a selected societal challenge (4 weeks). A formative assessment will happen mid-way through the assignment.	

Assessment 2: Individual assignment on developing and visualizing a concept based on the future context and design statement (4 -5 weeks), with a formative assessment after about 2 weeks. In addition, this assignment has an individual component which includes the development of a personal positioning essay within the design field and within society.

Assessment 3: Group assignment on developing and testing an experiential prototype and anticipating and visualising individual and social implications of the proposed concept.

Assignment 1: 40%

Assignment 2: 30%

Assignment 3: 30%

Students who score a course grade between 5,0 and 5,9 (with each separate assignment a 5,0 or higher) can complete a compensation assignment to repair the grade at the end of the semester (after assignment 4). The maximum grade for the course with a compensated assignment is 6,0.

Enrolment / Application

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Registration periods and deadlines are communicated on the website. See <http://www.io.tudelft.nl/osiris> for exact deadlines and details.

Registration beyond the registration deadlines is not possible.

Special Information

Contact: dp4-io@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Pre-Master Industrial Design Engineering

Select two

IOB4-B1-22	Strategic Brand Management	5
Responsible Instructor	Ir. S. Wu	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Brands define and differentiate organisations, drive performance and even shape the marketplace. In this elective students will be introduced to the fundamental concepts in brands and branding. They will find out how brand managers cultivate a brand position that resonates with consumers and sets them apart from competition, how brands can inform innovation and how design brings value to the process of branding. They will learn about the strategic and tactical decisions brand managers face in building and managing brands to maintain relevance and fuel growth.	
Study Goals	Upon finishing this course students are able to: LO1. Determine and motivate the positioning of a new brand in relation to opportunities in the market (extending LO2.9.1). LO2. Select, motivate, and develop brand touchpoints to realise the desired brand positioning for a new brand (extending LO2.10.3). LO3. Appraise possible management and innovation options for the brand to propose, motivate and elaborate a meaningful way to fuel brand relevance and growth (extending LO2.9.4).	
Education Method	The course consist of weekly lectures and tutorials, and a project assignment on which students will work as a group (around 4) for the duration of the course. Lectures introduce the theoretical material. Tutorials are a mix of studio discussions and exercises (to promote deeper understanding of the theoretical material and illustrate real world applications of theoretical content), and group feedback meetings and presentations (see Assessment). The brand building project entails the application of course material in the context of a new brand.	
Literature and Study Materials	Books or book chapters, academic/practitioner articles, online learning sources, reports, case texts. Full list of readings and access information will be available on Brightspace.	
Assessment	The learning objectives of the course are assessed through the brand building project report (group; due end of course). The report grade will be adjusted (up or down) for individual contribution based on peer assessment of group members contribution to the process and end result. Students will receive formative feedback on their brand building assignment at the group presentations and feedback sessions. Midway through the course, groups will be offered the opportunity to submit a draft report, should they wish to receive more targeted feedback.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: sbm-io@tudelft.nl	

IOB4-B2	Business Model Design	5
Course Coordinator	Dr. M. Bos-de Vos	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The Business Model Design elective is a course on business models specifically developed for design students. It provides bachelor students with an understanding of what business models are, how they can be designed and what the challenges and opportunities for business model design are. Starting with business models at the firm level (Part 1), the course will gradually move to business model design from the perspective of an interorganizational collaboration (Part 2), and an entire ecosystem (Part 3). Through working on a real-life business model design challenge of a case company from practice, students will acquire new knowledge, methods and skills that allow them to meaningfully engage in business model design activities in diverse situations, including new ventures, creative studios and networks of organizations. The business model design knowledge, methods and skills will be instrumental for the projects that students will execute in following courses (such as design projects and master courses in strategic design) and for determining the role they wish to have as a (strategic) design professional.	
Study Goals	Upon finishing this course students are able to: 1. Recognize and explain BM definitions and components. 2. Distinguish between different business model archetypes (deepening LO2.10.2) 3. Construct clear arguments for choosing a particular business model design and business model design method. 4. Identify relevant components for a new business model design, and map the value flows to show what the value for each stakeholder is (deepening LO 2.10.1). 5. Construct a business model design for a firm, an interorganizational collaboration, and ecosystem using existing methods. 6. Reflect on critical aspects of business model designs and formulate a statement that addresses or solves a business model design challenge in a particular context.	
Education Method	The course consists of a blended learning approach in which online materials, literature discussion groups, and business model design workshops are combined. The online materials (course information, literature, videos) are provided at the start of the course and can be studied in a self-paced manner. The literature discussion groups are held every two weeks (week 3.2, 3.4, 3.6). On Tuesdays, students discuss and work with the literature via an assignment that is provided in the studio that day. This serves as preparation for an in-depth plenary discussion with an expert and a business model design practitioner on Thursdays. The assignments and literature discussion groups allow students to work on study goals 1-3 and to prepare for the exam. Students need to come well-prepared on both days to have meaningful interactions. The business model design workshops are held in the other weeks (week 3.3, 3.5, 3.7) and focus on an existing business model design challenge of a case company from practice. During each of the three workshops, students will work on the successive parts of their assignment. Students will start the workshop on Tuesdays within their group. The instructions for the workshop are provided online and in the studio that day. On Thursdays, the workshops will continue and an expert and a representative of the case company will be available to provide formative feedback on the designs that are made. Afterwards, students will reflect on the results of their workshop. The workshops allow students to bring into practice what they have learned from the literature discussions, while working on study goals 4-6. They also help students to complete the full design assignment. All students have the opportunity to present or debate a selection of Business Model Designs from the assignment with the case company at the end of the course in a dragons den format.	
Literature and Study Materials	Mandatory literature & study materials: - Academic book chapters and articles (per topic). - Videos and materials on business model design methodology. For the literature discussions, students can study one of the provided documents per week, which is mandatory. Through sharing summaries of the articles within their group, working on the assignments with the group, and actively participating in the plenary discussion sessions with experts, students will also learn about the other documents. The exam questions are focused on specific business model design topics and can be answered by having read different sources.	
Assessment	The learning objectives of the course are assessed through an individual exam and an assignment. The individual exam will consist of multiple choice and open questions and will be 50% of the grade. The assignment consists of four parts: the three business model designs that were developed during the workshops including argumentation (these parts will be delivered with a group), and a business model design statement. The fourth part of the assignment can be developed individually or in a group with maximum 3 students. The full assignment accounts for 50% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: bmd-io@tudelft.nl	

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is structured in two pillars. In the first one, students learn through lectures, self-study, and group work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students also have lectures by external guests who provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. The second pillar of the course is about organising codesign sessions to tackle real-world challenges of service organisations. Students have the opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. Through practical assignments, students assume full responsibility for planning and executing codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided into groups, and each group will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.) and (2) identify opportunities to apply service codesign methods to create new services. The outcome of this work will be presented in a report that is collaboratively written by team members. Groups will receive feedback from their coaches on an initial draft of the report. This feedback is a formative assessment and does not count towards the final grade for the course. Then, after the delivery of this first report, coaches will establish a grade for assignment 1. This summative assessment represents 30% of a student's final grade. In assignment 2, the groups assembled for the first assignment will plan and execute codesign sessions to address one of the challenges identified before. This work will include a pilot session, which can be done with other students following the course, and another final session with external participants representing the target group. After receiving formative feedback from coaches, as done in the first assignment, groups will again deliver a report with the work done for assignment 2. An additional component of this report is a self-reflection on learning experiences done individually by each student in the group. When evaluating this second report, coaches can add or subtract a maximum of 1,0 points from the grade given the group's work in order to arrive at an individual grade for each student for assignment 2. This grade is summative and will represent 70% of students' final grade for the course.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cis-io@tudelft.nl	

IOB4-S2-24	Leadership and Project Organisation	5
Course Coordinator	Dr.ir. S.M. Flipse	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Leadership and Project Organisation is part of the fourth semester of the Bachelor program of Industrial Design Engineering (year 2, quarter 4). The intention of the course is to empower you as a designer to get your ideas across to your target audience and collaborators. For that, apart from being a good designer, you need to be able to plan, organise, lead and collaborate with others. The main desired outcome of the course is that you are able to apply insights and skills on project management, leadership and collaboration in a practical design setting.	
Study Goals	By the end of this course, you will be able to: LO1. discuss the relevance and significance of theoretical insights on leadership and project organization for designers; LO2. apply theoretical and practical insights on leadership, project organization and collaboration in an educational context; LO3. demonstrate that you are capable of effectively collaborating on a design project within a student team, while following a provided design project organization approach; LO4. write an individual learning report in which you demonstrate reflexive awareness of your personal learning process during this course.	
Education Method	You will learn by reading literature, attending interactive lectures, participating actively in workshops and practicing within and outside the walls of the course. The overall philosophy is that you learn skills best from the perspective of teaching these to others. Therefore you will design two educational modules, on project management, leadership and collaboration, for designers.	
Literature and Study Materials	Book chapters, (academic) articles, websites and other resources, made available via BrightSpace.	
Assessment	This course contains formative and summative assessment elements, carried out by peer students, coaches and the course coordinator. The course contains three main assignments. (1) a group assignment in which teams of students develop an interactive learning experience for peers, regarding leadership, project management and collaboration. (2) Individual assignments on theoretical and practical materials provided within the course. (3) individual reflection assignments on the student's individual learning process. Earlier versions of these assignments may be sent to coaches for formative assessment (feedback). Deadlines are communicated through BrightSpace.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	

IOB4-S3	Mastering Research Methods	5
Course Coordinator	E.A. van den Hende	
Contact Hours / Week x/x/x/x	2,5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course the student will collaborate with experienced researchers of our faculty on mastering scientific research in design and beyond. The student will discuss published research projects with them, and walk in their shoes during an academic research project Mastering research methods will help the student to develop scientific inquiry skills that are crucial for a plethora of if not all - courses at a university. In any (design) project, it is useful to be able to assess existing knowledge, and decide on the need for gathering new knowledge through research. Moreover, input for solid argumentation typically comes from existing or new research. Knowledge of strengths and shortcomings of research methods is key to stand ones ground. The course offers hands on research experience beyond creating merely awareness. With the help of experienced researchers, this course provides armor against scientifically proven claims, triggers curiosity into the unknown, and shows the merits of scientific research.	
Study Goals	Upon finishing this course students are able to: 1. evaluate research in terms of theoretical and practical implications, limitations, and generalizability (deepening LO 4.1.1, LO 4.3.1) 2. set-up research to conduct an empirical study for answering a research question that matches the topic and practical considerations laid out by the supervisor, and analyze the research accordingly (deepening LO 4.2.1, LO4.2.2, LO4.2.3) 3. present and report research and its outcomes adequately (deepening LO 4.3.2)	
Education Method	The course consists of a kick-off lecture, research expert discussions and an academic research project. The mandatory introduction lecture provides a walk-through of the course and introduces research method basics. The course further includes discussion meetings with research experts to further understanding of quantitative research. These discussion meetings require preparations by reading and practicing with the self-study materials. In parallel runs an academic research project, executed in duos in close collaboration with a research expert. The research experts formulate these studio projects and set the conditions. Students can indicate their preferences for a specific studio project theme (upon availability) and formulate their focus within this project. Depending on the focal topic, and desired efficiency and peer support, the first part of the research project is a collaborative effort: one or multiple duos discuss research topics and designs and contribute to a project database that will be available to all for further analyses. The research expert supervises a specific studio from the start until the final deliverable. The research expert facilitates regular meetings to discuss project progress, and provides suggestions and support regarding project planning and data analysis.	
Literature and Study Materials	Statistics self-study materials on Brightspace (Stats 1, Stats 2, Stats 3) Specific academic papers, depending on the participating research experts in the course. We will detail a reading list one week prior to the course starts and ensure that all can be downloaded from the library and/or are accessible on Brightspace. Literature (for reference only): Kumar, R. (2019). Research methodology: A step-by-step guide for beginners. Sage. (fourth edition is also suitable)	
Assessment	The course includes summative assessment of the research paper (written by duos) about the academic research project. Formative / peer feedback prepares for this assessment in two ways: feedback on the statistics self-study, and on the parts of the research paper.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mrm-io@tudelft.nl	

IOB6-E8	Design Analytics	5
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period	Dr. H. Verma 4 3 3 3 4	
Course Language	English	
Course Contents	In this technological and methodological elective, students will gather tools and acquire knowledge to critically evaluate (interactive and intelligent) technological prototypes and set the stage for the next iteration of the design phase by eliciting actionable design guidelines based on comprehensive empirical analysis. The technological component of this course is demonstrated by the strong focus on the improvement of IPSS (intelligent products, services, and systems) such as, conversational assistants, wearables, etc. Instead of examining the complete design cycle, students will focus solely on the quantitative assessment of IPSS and examine their cognitive, perceptual, and experiential effects. To this end, students will learn and add skills to their methodological toolkit, which will enable them to a) design sound experiments (with Control and Experimental groups), b) use sensing technologies (e.g., eye-tracking) and adapt existing data-acquisition tools (e.g., NASA TLX, MSLQ), c) identify the right statistical tests (descriptive vs. inferential; comparison vs. regression), d) discriminate correlation from causation and identify confounding factors, and e) understand the proper means of communicating their quantitative analysis and results. The content of this course is relevant for students because they learn how to systematically assess their designs with standardized practices and communicate their findings with multiple stakeholders and a broader design-research community. In this way, this course offers an essential ingredient in students transformation from a UX Designer to a UX Researcher. In this course, the student will become familiar with the 5-stage Design Analytics Process (DAP) i.e., design diagnostics, study design synthesis, experimentation, data analysis, and design prescription.	
Study Goals	Upon finishing this course, students are able to: 1: Describe the design unknowns or flaws of a given product, service, or system (deepening LO 1.3.3, LO 1.6.3). 2: Formulate research questions and pertinent hypotheses to empirically evaluate the given technology (deepening LO 4.2.1). 3: Design an experiment protocol to validate the research hypotheses (deepening LO 4.2.2, LO 1.6.1). 4a: Evaluate the quality of collected multi-modal data (deepening LO 2.4.4). 4b: Apply the pre-designed experiment protocol with real subjects (deepening LO 1.6.1) 5: Analyze experimental data using appropriate statistical methods (deepening LO 2.1.1). 6: Explain the results of statistical analysis for future design improvement (deepening LO 4.3.1, LO 4.3.2).	
Education Method	This 10-week course consists of 8 weekly lectures (2h/week), 8 weekly hands-on workshops (upto 1.5h/week) with Teaching Assistants, and 8 weekly thematic coaching sessions (0.5h/week). Students are expected to prepare themselves for an upcoming lecture by studying the provided reading material and by watching the provided educational videos in a flipped classroom format. Lectures and hands-on-sessions will be related in terms of content to ensure that the theory--will be put into practice--in the hands-on workshops. Moreover, student groups will have weekly discussions with thematic coaches, who will guide them in their particular design briefs. Inherently, this course provides the ideal opportunity for applying an inquiry-based learning approach at scale. As such, we will invite colleagues, interested parties, and stakeholders to submit their IPSS (Intelligent Products, Services, and Systems) designs for evaluation a few months before the start of the course. Group assignments In week 9, the student groups are expected to submit a comprehensive report documenting their engagement with the 5-stage Design Analytics Process, and their empirical evaluation of the chosen IPSS. Weekly sessions with the thematic coach (formative assessments) will allow groups the possibility of seeking feedback on their project direction and progress. Kickstarting lecture and hands-on session (week 1) At the kickstarting lecture, we will employ ice-breaking techniques (e.g., "speed-dating") for enabling students to familiarize among themselves. Next, we will perform a brief diagnostic assessment by using the Kahoot! game-based learning platform. Students will interactively answer questions that determine their personality profiles (e.g., the Big-5), interests, and grouping preferences so that we form effective and diverse student teams (3--4 persons). After a brief presentation of all the available IPSS designs, the newly formed teams will bid for the design they prefer to evaluate throughout this course by delivering a final report at the end of the course. Intermediate lectures and hands-on sessions (weeks 2--8) In each lecture, we will showcase the research and experimental thinking that corresponds to the given stage in the course, and in full alignment with the established LOs. To this end, we will present examples of existing IPSS designs tested through our 5-stage DAP, outlined above, in the form of "evaluation cases": An evaluation case can be a prototype borrowed from literature where we describe how specific design unknowns are targeted for formulating concrete RQs/Hypotheses. During lectures, we will define our threshold concepts--notions that once understood change the perception of a subject, phenomenon, or experience. During the lectures, we will use Kahoot! for increasing engagement with brief quizzes and for collecting learning feedback in situ. In the following hands-on sessions, the comprehension of the threshold concepts will be tested with practical work on other evaluation cases. Then, the students will plan how they will move forward with their evaluation cases in collaboration with the thematic coaches.	
Literature and Study Materials	The course coordinators will design a reader on Brightspace with a curated set of readings, lecture notes, web resources, hands-on tutorials, and videos for different modules within this elective. The mandatory weekly readings will be selected chapters from a list of relevant books, and other articles.	
Assessment	The learning objectives of this elective will be assessed through a group report, including a final presentation, and an individual exam. Group deliverables will contribute to 60% of the students final grade. The final exam will contribute to the remaining 40%. Group Work (60%): Student groups will deliver a final report at the end of the elective (max. 20 pages), accompanied by a final presentation. This group deliverable corresponds to the aforementioned DAP stages and the underlying LOs. The LOs are fulfilled by the rigorous evaluation of an IPSS (design artifact), for which the groups will work throughout the duration of the course. The course coordinators will provide a template and assessment rubric for these reports, and instructions on the length and structure of each report. The final report will manifest into a research portfolio that will be presented publicly at the end of the course to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. The final individual grade for the group work will be the sum of the grades for each group report, weighted by the peer contribution assessment scores produced weekly.	

Enrolment / Application	Individual Exam (40%): A final exam will constitute a mix of MCQs (multiple choice questions) and open questions. Formative Assessment: Weekly thematic coaching will provide students with the necessary feedback as well as enable them to assess the progress of their learning, their conceptual understanding. In addition, thematic coaching will assist students to effectively align their project direction by reflecting critically about the diverse aspects of design analytics. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible
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